CONTINUOUS AND SMOOTH PLURISUBHARMONIC EXHAUSTIONS: THREEFOLD COUNTEREXAMPLES AND PARTIAL RESULTS FOR SURFACES

Abstract. Starting from Koike's modified Grauert example, we construct a smooth quasi-projective Kähler threefold with a locally Lipschitz plurisubharmonic exhaustion but no C^2 plurisubharmonic exhaustion. A modification over one point gives a non-Kähler threefold with the same properties. For divisor complements in compact complex surfaces, we prove that continuous and smooth plurisubharmonic exhaustions are equivalent to a boundary intersection-matrix condition in two settings: every nonprojective compactifier, without a Kähler assumption, and every smooth $\mathbf{P}^1$-bundle with reduced simple normal crossing boundary, without a degree restriction. Further affirmative results treat disconnected projective boundaries and several log surface cases. For smooth square-zero boundaries, character extension and irregularity give sufficient conditions independent of continuous weak completeness. Scoped holonomy arguments sharpen the remaining neighborhood problem. The unrestricted implication in dimension two, including the general Kähler and quasi-projective cases, is not proved.

1. Introduction and main results

The regularity of a plurisubharmonic exhaustion matters even for Kähler manifolds. This paper constructs a smooth quasi-projective threefold admitting a continuous plurisubharmonic exhaustion but no C^2 one, and then inserts a non-Kähler configuration over a single point without changing these exhaustion properties. In dimension two, the same line-bundle construction cannot give a counterexample. We prove this obstruction and several affirmative regularization results for surfaces, and specify the boundary problem left by our reductions. Throughout the introduction, "psh" abbreviates plurisubharmonic, and "smooth" means C^∞.

1.1. **Threefold counterexamples and their metric construction.** Our compact base is the modified Grauert ruled surface appearing in Koike's work [1, Example 4.1]. Its failure of semipositivity is already known; the first-order obstruction, including its C^2 version, is not claimed as a new result here [1, Theorem 1.4 and Example 4.1], [2, Remark 2.8]. We include a direct proof to make the regularity threshold used below explicit. The analytic construction

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then consists of a cusp correction, a global maximum, and a radius criterion converting any hypothetical C^2 exhaustion into a forbidden metric. Failure of smooth semipositivity by itself would not provide a continuous exhaustion.

The cusp estimate also reaches the locally Lipschitz endpoint: a sufficiently large multiple of $q^{1/2}$ dominates the initial curvature in the scaled tangent variables. The coefficient is essential at this endpoint; for smaller exponents shrinking the neighborhood already suffices.

Theorem 1.1. *Let C be the smooth projective completion of*

$$y^2 = x^5 - x,$$

let $K = K_C$, and let

$$0 \longrightarrow K \longrightarrow V \longrightarrow \mathcal{O}_C \longrightarrow 0 \tag{1.1}$$

be a nonsplit holomorphic extension. Let

$$S = \mathbf{P}_{\text{lines}}(V), \qquad \pi : S \longrightarrow C, \qquad Y = \mathbf{P}(K) \subset S,$$

and set

$$L = \mathcal{O}_S(Y) \otimes \pi^* K, \qquad M = \operatorname{Tot}(L^{-1}). \tag{1.2}$$

Then:

(i) *S is a smooth projective surface and M is a smooth quasi-projective Kähler threefold;*
(ii) *L admits a continuous Hermitian metric with plurisubharmonic local weights, but admits no C^2 Hermitian metric with semipositive curvature;*
(iii) *M admits a continuous plurisubharmonic exhaustion, but admits no C^2 plurisubharmonic exhaustion.*

The metric and the exhaustion in (ii)–(iii) *can be chosen locally Lipschitz, and hence locally of class $C^{0,\gamma}$ for every $0 < \gamma \leq 1$. No assertion about existence of a C^1 exhaustion is made.*

The passage from metrics to exhaustions uses a criterion valid for a holomorphic line bundle A over any compact complex manifold Z:

$$\operatorname{Tot}(A^{-1}) \text{ admits a } C^r \text{ psh exhaustion}$$
$$\Longleftrightarrow$$
$$A \text{ admits a } C^r \text{ semipositive Hermitian metric}, \qquad r \in \{2, \infty\}.$$

This is Corollary 3.4. A semipositive metric gives the squared dual norm exhaustion. Conversely, averaging an arbitrary exhaustion under fiberwise circle rotation and taking a sufficiently high level yields a radius function whose negative logarithm is a psh metric weight. Consequently, the obstruction in Theorem 1.1 rules out every C^2 psh exhaustion, not only a smoothing of the particular one constructed here.

The non-Kähler example has the same dimension and agrees with M away from a compact exceptional fiber.

Theorem 1.2. *For every point x of the threefold M in Theorem 1.1, there is a smooth non-Kähler threefold N and a proper holomorphic modification $\mu : N \to M$ inducing a biholomorphism*

$$N \setminus \mu^{-1}(x) \simeq M \setminus \{x\}.$$

The threefold N admits a locally Lipschitz psh exhaustion but no C^2 psh exhaustion.

The construction places Hironaka's opposite-order blowups inside the exceptional plane of the blowup of x. It produces two compact rational curves whose sum is zero in real homology, precluding a Kähler form. The exhaustion assertions follow from Proposition 4.1: a proper surjective holomorphic map that is biholomorphic away from a compact subset preserves the existence of C^r psh exhaustions, for $r = 0, 2, \infty$. The reverse implication uses convex composition to make an exhaustion constant near the exceptional set before descending it. Thus the modification is localized over one point; no small deformation of a fixed complex structure is asserted. Choosing x outside the zero section also preserves a neighborhood of the compact base. The compact product principle in Proposition 4.4 gives further examples in higher dimensions, including the Hopf-surface product in Corollary 4.5.

1.2. **Surface results and the remaining boundary problem.** For a line bundle A over a compact connected Riemann surface B, the existence of a continuous psh exhaustion on $\mathrm{Tot}(A^{-1})$ is equivalent both to $\deg A \geq 0$ and to the existence of a smooth psh exhaustion (Corollary 5.3). Thus the threefold construction cannot be lowered in dimension by replacing its compact base with a curve.

Holomorphic convexity gives a different affirmative mechanism. Vâjâitu's theorem [6] implies that a connected noncompact complex surface with a continuous psh exhaustion and a nonconstant holomorphic function admits a smooth psh exhaustion (Proposition 5.4). A criterion of Mongodi–Slodkowski also applies when an exhaustion takes arbitrarily large values outside the image of the continuous minimal kernel (Proposition 5.5). Neither statement requires a compactification. Their separate ambient real-analytic regularization theorem has an additional hypothesis, recorded in Subsection 5.2.

Our principal geometric results concern divisor complements $X = S \setminus \operatorname{supp} D$, where S is a smooth compact complex surface and $D \neq 0$ is effective. The existence of this compactification is an additional assumption. Only the support of D determines X.

Theorem 1.3 (Two numerical criteria for surface exhaustions)**.** *Let S be a connected smooth compact complex surface, let $D \neq 0$ be effective, and put $X = S \setminus \operatorname{supp} D$. Suppose that either*

(i) *S is nonprojective, with no Kähler or normal-crossing assumption; or*
(ii) *$S \to B$ is a smooth $\mathbf{P}^1$-bundle over a compact connected Riemann surface and D is reduced with simple normal crossings.*

Then the following are equivalent:

(a) *X admits a continuous psh exhaustion;*
(b) *X admits a smooth psh exhaustion;*
(c) *no connected component of* $\operatorname{supp} D$ *has negative definite intersection matrix.*

In (ii)*, there is no restriction on the boundary degree or the number of connected boundary components.*

The two cases are proved in Theorem 5.15 and Corollary 6.9. The negative definite obstruction is local: contraction to a normal surface point contradicts the Hartogs maximum principle for a divergent psh function. For a nonprojective surface of algebraic dimension one, the matrix condition forces the boundary to be a union of whole elliptic fiber supports. In algebraic dimension zero, reduction to a minimal model and Enoki's theorem handle the class VII case; the remaining Hopf boundaries are smooth elliptic curves. The elliptic-neighborhood step uses the September 2026 preprint [16]. In the smooth ruled case, a connected square-zero boundary is either a fiber or a smooth étale multisection. A finite étale base change reduces the latter to an affine-line bundle, a punctured degree-zero line bundle, or a product with a punctured projective line.

For projective compactifiers, disconnected boundary support also implies the continuous-to-smooth conclusion (Theorem 5.14). Resolution with the full reduced boundary preserves the open surface. The intersection matrix gives smooth strictly psh collars in the positive-eigenvalue case. The residual projective case has a connected reduced simple normal crossing boundary supporting a divisor

$$A = \sum_i a_i D_i, \quad a_i > 0, \qquad A \text{ nef}, \quad A^2 = 0, \quad A \cdot D_i = 0,$$

with $\kappa(S, A) = 0$ and $K_S \cdot A \geq 0$ (Corollary 5.20). The continuous exhaustion hypothesis is retained in this reduction.

We also prove the implication for smooth log Fano pairs, connected numerically log Calabi–Yau pairs, and elliptic fibrations with vertical boundary (Propositions 6.11, 6.12, and 6.14). These statements concern the indicated smooth compactifications; arbitrary log minimal model steps need not preserve the complement.

For a smooth connected curve C with $C^2 = 0$ in a compact Kähler surface, Ueda theory supplies further sufficient conditions without assuming continuous weak completeness (Proposition 6.3). Let ρ be the unitary character of $N_{C/S}$ and put

$$K = \ker\bigl(H_1(C, \mathbb{Z}) \longrightarrow H_1(S, \mathbb{Z})\bigr).$$

If $\rho(K)$ is finite, a positive power of the character extends to S, and $S \setminus C$ has a smooth psh exhaustion (Theorem 6.5). This holds in particular when the homology map is injective, or when $q(S) = h^1(S, \mathcal{O}_S) \geq g(C)$ (Corollary 6.6).

The examples in Section 7 separate these questions from holomorphic convexity and from the log Kodaira dimension. Some complements have smooth psh exhaustions and only constant holomorphic functions even when the ordinary canonical bundle is ample. Other pairs realize arbitrary degree-zero normal bundles; no continuous exhaustion is asserted for their arbitrary parameters.

The residual analytic problem already includes smooth connected square-zero curves of genus at least two. Any counterexample of this form must have $q(S) < g(C)$ and

$$\operatorname{rank} K = 2(g(C) - q(S)), \qquad \rho(K) \text{ infinite},$$

as well as an infinite-type neighborhood which is not unitary flat (Corollary 8.5). A continuous collar which is constant on the leaves of a given holomorphic foliation does force unitary linearization; a nonlinearizable irrational cyclic holomorphic suspension admits no continuous divergent psh collar (Propositions 8.1 and 8.2). These statements do not produce a foliation or leafwise constancy from an arbitrary psh collar. The full continuous-to-smooth implication for complex surfaces, including all Kähler surfaces and all projective divisor complements, remains unproved here.

Sections 2–4 give the detailed threefold constructions. Section 5 establishes the surface reductions and the nonprojective criterion. Section 6 develops the homological, ruled, and log surface results. Section 7 gives the examples, and Section 8 places the holonomy obstructions beside the precise remaining problem.

1.3. **Relation to prior work and scope of the claims.** The known ruled-surface obstruction, Hironaka's opposite-order blowups, and the quoted classification and Ueda theorems are inputs, not new constructions of this paper. The nonconstant-holomorphic-function and minimal-kernel criteria below are explicitly recorded consequences of Vâjâitu and Mongodi–Slodkowski. Likewise, the affine-line-bundle step in the smooth-ruling argument reproduces a construction of Koike–Ueda. The surface results combine these inputs with the boundary and character arguments given here; we do not claim priority for every intermediate criterion or for a general regularization theorem.

The self-contained threefold argument is the combination of the explicit bounded-potential metric and the radius obstruction, followed by a compactly supported modification. Remark 2.5 explains exactly which part of the bounded-potential question in [3, Section 4] this construction addresses. We make no claim that dimension three is the smallest possible dimension of a counterexample, nor that Lipschitz regularity is an optimal threshold. The full dimension-two problem is not settled in this article.

Some surface conclusions use Theorem 1.6 and Proposition 6.2(B) of the September 2026 preprint [16]; these uses are stated at the relevant steps and

are independent of the threefold construction. The October 2026 preprint [19] concerns a different neighborhood condition, as explained in Remark 8.4.

1.4. **Conventions and terminology.** We use

$$d^c = \frac{i}{4\pi}(\bar\partial - \partial), \qquad dd^c = \frac{i}{2\pi}\partial\bar\partial.$$

For a holomorphic line bundle A and a local holomorphic frame e, a Hermitian metric is written

$$\|e\|_h^2 = e^{-\varphi}.$$

Its curvature is $\Theta_h(A) = dd^c\varphi$, representing $c_1(A)$. If $e_j = g_{ij}e_i$, the metric weights satisfy

$$\varphi_j = \varphi_i - \log|g_{ij}|^2. \tag{1.3}$$

A positive continuous Hermitian metric is semipositive in the current sense when its local weights are continuous psh functions. For a C^2 metric this agrees with pointwise semipositivity of its curvature.

For a C^2 real-valued function f, its unnormalized Levi form is

$$\mathcal{L}_f(\xi) = \sum_{j,k} f_{z_j\bar z_k}\xi_j\overline{\xi_k}.$$

Thus f is psh exactly when $\mathcal{L}_f \geq 0$. The fixed positive factor $1/(2\pi)$ in dd^c does not affect any positivity argument. Poincaré–Lelong takes the form

$$dd^c \log\|s_D\|_h^2 = [D] - \Theta_h(\mathcal{O}(D)), \tag{1.4}$$

where s_D is the canonical section of an effective divisor D.

Definition 1.4. A continuous real-valued function F on a complex manifold X is an *exhaustion* if every sublevel set $\{F \leq c\}$ is compact. Equivalently, $F(x_j) \to +\infty$ for every sequence escaping every compact subset of X.

We say that X is *continuously weakly complete* or *smoothly weakly complete* when it admits a continuous or smooth psh exhaustion, respectively. Specifying the regularity avoids the varying conventions for weak completeness in the literature. Our projectivization $\mathbf{P}_{\text{lines}}(E)$ parametrizes one-dimensional subspaces of the fibers of E, and its tautological line bundle is $\mathcal{O}_{\mathbf{P}(E)}(-1)$.

2. The ruled surface and continuous semipositive metrics

2.1. The ruled surface and the metric obstruction.

2.1.1. *The curve and the extension.* The polynomial $x^5 - x$ has five distinct roots. The double covering $C \to \mathbf{P}^1$ is therefore branched at these five roots and at infinity. Riemann–Hurwitz gives

$$2g(C) - 2 = 2(-2) + 6 = 2,$$

so $g(C) = 2$ and $\deg K = 2$. Serre duality gives

$$\operatorname{Ext}^1(\mathcal{O}_C, K) = H^1(C, K) \simeq H^0(C, \mathcal{O}_C)^* \simeq \mathbb{C}.$$

Consequently a nonzero extension class exists. All nonzero classes give isomorphic extensions after allowing scalar automorphisms of the end terms, so the isomorphism class of V is unambiguous.

One can also specify the extension through a Dolbeault operator. Choose a smooth positive $(1,1)$-form β on C with $\int_C \beta = 1$, and regard it as a K-valued $(0,1)$-form. On the smooth bundle $K \oplus \mathcal{O}_C$, set

$$\bar\partial_V(s,f) = (\bar\partial_K s + f\beta, \bar\partial f). \tag{2.1}$$

This operator is integrable, since there are no $(0,2)$-forms on a curve, and defines (1.1). A splitting would give a holomorphic lift $(s,1)$ and hence $\bar\partial_K s = -\beta$. Since s is a smooth $(1,0)$-form, Stokes' theorem would then imply $\int_C \beta = 0$, a contradiction. The choice of the standard identification of K-valued $(0,1)$-forms with $(1,1)$-forms only changes this argument by a sign.

2.1.2. *The negative section and the line bundle.* By GAGA, V is algebraic, so $S = \mathbf{P}_{\text{lines}}(V)$ is a smooth projective surface. The inclusion $K \hookrightarrow V$ gives a section $Y = \mathbf{P}(K)$ of π. Its normal bundle is

$$N_{Y/S} \simeq \operatorname{Hom}(K, V/K) \simeq K^{-1}. \tag{2.2}$$

In particular,

$$\mathcal{O}_S(Y)|_Y \simeq K^{-1}, \qquad L|_Y \simeq \mathcal{O}_Y, \qquad Y^2 = -2. \tag{2.3}$$

The trivialization of $L|_Y$ in (2.3) is the canonical one induced by $K^{-1} \otimes K$.

For context, this line bundle is nef and big. If F is a fiber of π, then

$$L \cdot Y = 0, \qquad L \cdot F = 1, \qquad L^2 = 2.$$

For every irreducible curve $D \neq Y$ dominating C,

$$L \cdot D = Y \cdot D + 2\deg(\pi|_D) > 0.$$

Every vertical irreducible curve is a fiber. Hence L is nef, and the usual surface criterion $L^2 > 0$ implies bigness. For completeness, the latter also follows from Riemann–Roch: $h^2(S, L^m) = 0$ for m large, because $(K_S - mL)$ has negative intersection with a fixed ample divisor, and therefore

$$h^0(S, L^m) \geq \chi(S, L^m) = m^2 + O(m).$$

Moreover, $2L - Y = Y + 2\pi^* K$ is ample by Nakai–Moishezon: its square is 6, its intersections with Y and a fiber are 2 and 1, and its intersection with every other irreducible curve is positive.

2.1.3. *First-order transition functions.* Choose a finite coordinate cover $\{U_i\}$ of C, frames k_i of K, and holomorphic lifts v_i of 1 in (1.1). On overlaps,

$$k_j = g_{ij} k_i, \qquad v_j = v_i + c_{ij} k_i. \tag{2.4}$$

The cocycle $\{c_{ij} k_i\}$ represents the nonzero extension class. Near Y, write a line in V as

$$[k_i + w_i v_i].$$

Expressing this line in the j-th frame yields

$$w_j = \frac{g_{ij} w_i}{1 - c_{ij} w_i}. \tag{2.5}$$

Shrinking the neighborhoods if necessary, all denominators are nonzero. Let d_i be the frame of $\mathcal{O}_S(Y)$ defined by $s_Y = w_i d_i$. Then

$$\frac{d_j}{d_i} = \frac{w_i}{w_j} = \frac{1 - c_{ij} w_i}{g_{ij}}.$$

Thus the frames $\ell_i = d_i \otimes \pi^* k_i$ of L satisfy

$$\ell_j = (1 - c_{ij} w_i) \ell_i. \tag{2.6}$$

Their restrictions to Y agree, in accordance with the trivialization in (2.3).

Proposition 2.1. *The line bundle L admits no C^2 Hermitian metric with semipositive curvature.*

Proof. Suppose such a metric exists, with C^2 psh weights φ_i in the frames ℓ_i. By (2.6), the functions $\varphi_i|_Y$ agree on overlaps. They define a subharmonic function on the compact connected curve Y, hence a constant. Rescaling the metric, we may assume

$$\varphi_i(z, 0) = 0. \tag{2.7}$$

In particular $\varphi_{i, z\bar{z}}(z, 0) = 0$. The Levi matrix at $(z, 0)$ is positive semidefinite. A positive semidefinite Hermitian matrix with a zero diagonal entry has zero entries in the corresponding row and column. Therefore

$$\varphi_{i, w\bar{z}}(z, 0) = 0.$$

The C^1 function

$$a_i(z) = \varphi_{i, w}(z, 0)$$

satisfies the Cauchy–Riemann equation and is holomorphic.

Equations (1.3) and (2.6) give

$$\varphi_j(z, w_j) = \varphi_i(z, w_i) - \log |1 - c_{ij}(z) w_i|^2,$$

where the base coordinate change is understood. Differentiating with respect to w_i at zero and using (2.5), we obtain

$$g_{ij} a_j = a_i + c_{ij}. \tag{2.8}$$

The holomorphic lifts $v_i' = v_i - a_i k_i$ consequently agree on overlaps:

$$v_j' = v_i + c_{ij} k_i - a_j g_{ij} k_i = v_i - a_i k_i = v_i'.$$

They define a global holomorphic lift of 1, splitting (1.1). This is a contradiction. □

This calculation is the specialized first-order obstruction occurring in [1, Theorem 1.4 and Example 4.1]. Related obstruction classes, including their applicability to C^2 metrics, are discussed in [2]. Neither reference is needed for the preceding proof.

2.2. **A local continuous metric near a negative curve.** We isolate the analytic estimate used in the construction. The unitary flatness assumption means that a smooth metric of zero curvature exists on the restricted line bundle.

Lemma 2.2 (Cusp correction)**.** *Let D be a compact connected smooth curve in a complex surface X. Suppose $\deg N_{D/X} < 0$, and let A be a holomorphic line bundle such that $A|_D$ is unitary flat. Choose a smooth metric h_0 on A restricting to a flat metric on D, and a smooth metric h_D on $\mathcal{O}_X(D)$ whose curvature is strictly negative on TD. Put*

$$q = \|s_D\|^2_{h_D}.$$

For every $0 < a \le \frac{1}{2}$, there is a constant $\Lambda > 0$ such that the local weights of h_0 plus Λq^a are strictly psh off D on a sufficiently small neighborhood of D and extend as continuous psh weights across D. They are locally of class $C^{0,2a}$; in particular they are locally Lipschitz when $a = \frac{1}{2}$. For $0 < a < \frac{1}{2}$ one may take $\Lambda = 1$ after shrinking the neighborhood. At $a = \frac{1}{2}$, a sufficiently large Λ suffices.

Proof. The chosen metrics exist. Negative degree on the curve gives a negative curvature metric on $N_{D/X} = \mathcal{O}_X(D)|_D$. Any metric on a restricted line bundle extends smoothly to a neighborhood: extend the logarithm of its ratio to a fixed smooth metric. The same argument extends the flat metric on $A|_D$.

Use finitely many relatively compact coordinates (z, w) near D, with $D = \{w = 0\}$. Write

$$q = |w|^2 e^{-b(z,w)}, \qquad \beta = -b,$$

where b is the smooth weight of h_D. Negativity along TD says

$$\beta_{z\bar{z}}(z, 0) > 0. \tag{2.9}$$

For $w \neq 0$, set $\tau = \log q = \log |w|^2 + \beta$. The chain rule gives the exact identity

$$\mathcal{L}_{q^a}(v) = q^a(a\mathcal{L}_\tau(v) + a^2|\partial\tau(v)|^2). \tag{2.10}$$

Evaluate this on

$$v = \xi\frac{\partial}{\partial z} + w\nu\frac{\partial}{\partial w}.$$

Since $\log |w|^2$ is pluriharmonic off D,

$$\mathcal{L}_\tau(v) = \beta_{z\bar{z}}(z, 0)|\xi|^2 + O(|w|)(|\xi|^2 + |\nu|^2),$$

and

$$\partial\tau(v) = \nu + \beta_z(z, 0)\xi + O(|w|)(|\xi| + |\nu|).$$

Therefore

$$\frac{\mathcal{L}_{q^a}(\xi, w\nu)}{q^a} = a\beta_{z\bar{z}}(z,0)|\xi|^2 + a^2|\nu + \beta_z(z,0)\xi|^2 + O(|w|)(|\xi|^2 + |\nu|^2). \tag{2.11}$$

All remainder bounds are uniform after choosing the finite cover and shrinking its coordinate neighborhoods.

The leading Hermitian form in (2.11) is positive definite: vanishing of its first term forces $\xi = 0$ by (2.9), and vanishing of the second then forces $\nu = 0$. Its smallest eigenvalue has a positive uniform lower bound on the compact coordinate pieces covering D. Also q^a is uniformly comparable to $|w|^{2a}$. Thus, after shrinking the neighborhoods,

$$\mathcal{L}_{q^a}(\xi, w\nu) \geq c|w|^{2a}(|\xi|^2 + |\nu|^2) \tag{2.12}$$

for a uniform $c > 0$.

Let φ_0 be a local weight of h_0. Its restriction to D has zero curvature, so $\varphi_{0,z\bar{z}}(z,0) = 0$. Smoothness gives

$$\varphi_{0,z\bar{z}} = O(|w|), \qquad \varphi_{0,z\bar{w}} = O(1), \qquad \varphi_{0,w\bar{w}} = O(1).$$

Consequently, in the same scaled variables,

$$|\mathcal{L}_{\varphi_0}(\xi, w\nu)| \leq C|w|(|\xi|^2 + |\nu|^2). \tag{2.13}$$

Combining (2.12) and (2.13) yields

$$\mathcal{L}_{\varphi_0 + \Lambda q^a}(\xi, w\nu) \geq (\Lambda c|w|^{2a} - C|w|)(|\xi|^2 + |\nu|^2). \tag{2.14}$$

If $0 < a < \frac{1}{2}$, take $\Lambda = 1$ and then shrink until the coefficient is positive. If $a = \frac{1}{2}$, fix $\Lambda > C/c$; the coefficient becomes $(\Lambda c - C)|w| > 0$ throughout the punctured neighborhood where the two uniform estimates hold. The constants c, C were obtained before choosing Λ, so this choice is not circular. Every tangent vector at $w \neq 0$ can be written in these scaled variables. This proves strict plurisubharmonicity off D in both cases.

The function $q^a = |w|^{2a}e^{-ab}$ is continuous and locally $C^{0,2a}$, including $2a = 1$. The local weights $\varphi_0 + \Lambda q^a$ are bounded near D and extend continuously to it. The removable singularity theorem for psh functions across an analytic hypersurface therefore proves that these extended weights are psh. Adding the globally defined function Λq^a preserves the metric transition law (1.3). □

Remark 2.3. The scale in (2.11) is essential. The normal component of the Levi form has order $|w|^{2a-2}$, while the tangential component has order $|w|^{2a}$. These two scales allow the cusp to dominate the bounded mixed curvature of a smooth initial metric. At $a = \frac{1}{2}$ the scaled cusp and the initial error both have order $|w|$, which explains the coefficient choice in (2.14). Ordinary smoothing of the cusp would not preserve this domination at D. The argument proves no C^1 regularity.

2.3. **An explicit global continuous metric.** Return to the ruled surface S and the line bundle L in (1.2). Choose a positive smooth metric h_K on K. Using (2.3), extend the dual metric h_K^{-1} on $\mathcal{O}_S(Y)|_Y$ to a smooth metric h_Y on $\mathcal{O}_S(Y)$. Set

$$h_0 = h_Y \otimes \pi^* h_K, \qquad \theta_0 = \Theta_{h_0}(L), \qquad q = \|s_Y\|^2_{h_Y}. \tag{2.15}$$

The metric $h_0|_Y$ is the flat metric under the canonical trivialization of $L|_Y$, and

$$\theta_0|_{TY} = 0, \qquad \Theta_{h_Y}(\mathcal{O}(Y))|_{TY} = -\Theta_{h_K}(K) < 0. \tag{2.16}$$

Fix $0 < a \le \frac{1}{2}$ and choose $\Lambda > 0$ as in Lemma 2.2. Since q vanishes exactly on Y and S is compact, there exists $\varepsilon > 0$ such that the weights $\varphi_0 + \Lambda q^a$ are psh on $\{q < \varepsilon\}$. Write $Q = \max_S q$ and choose $0 < \varepsilon < Q$. Choose a constant

$$B > \max_{\varepsilon \le t \le Q} (\Lambda t^a - \log t). \tag{2.17}$$

Define, on $[0, Q]$,

$$\chi(0) = 0, \qquad \chi(t) = \max\{\Lambda t^a, \log t + B\} \quad (t > 0), \tag{2.18}$$

and put

$$h = h_0 \exp(-\chi(q)). \tag{2.19}$$

Proposition 2.4. *The metric (2.19) is a positive continuous Hermitian metric on L with psh local weights. It is locally $C^{0,2a}$, and is locally Lipschitz for $a = \frac{1}{2}$.*

Proof. The logarithmic branch in (2.18) tends to $-\infty$ as $t \to 0$, so $\chi(t) = \Lambda t^a$ for all sufficiently small t. Hence χ is continuous at zero and bounded on $[0, Q]$.

On $S \setminus Y$, Poincaré–Lelong and (2.15) give

$$\theta_0 + dd^c \log q = \pi^* \Theta_{h_K}(K) \ge 0. \tag{2.20}$$

As a current on S, the same expression equals $\pi^*\Theta_{h_K}(K) + [Y]$. Thus $\varphi_0 + \log q + B$ is a singular psh weight of L.

On $\{0 < q < \varepsilon\}$, the weights of h are

$$\varphi_0 + \chi(q) = \max\{\varphi_0 + \Lambda q^a, \varphi_0 + \log q + B\},$$

a maximum of two psh functions. They are therefore psh. On $\{q \ge \varepsilon\}$, (2.17) makes the logarithmic branch strictly dominant. This domination also holds on a collar of $\{q = \varepsilon\}$, so no boundary gluing issue arises. The weights are psh there by (2.20). Near Y, the cusp branch is dominant, and Lemma 2.2 proves positivity across Y.

All the weights are finite and continuous. Near Y their only nonsmooth term is $\Lambda q^a = \Lambda |w|^{2a} e^{-ab}$, which is $C^{0,2a}$ for $0 < 2a \le 1$. Away from Y, both branches are smooth, and their maximum is locally Lipschitz, hence locally $C^{0,2a}$. This proves the asserted regularity. □

Remark 2.5 (Bounded potentials and minimal singularities). The current

$$T = \theta_0 + dd^c\chi(q) \in c_1(L)$$

has locally Lipschitz potentials when $a = \frac{1}{2}$. It also has minimal singularities: if $T' = \theta_0 + dd^c v \geq 0$ is another positive current in $c_1(L)$, its quasi-psh potential v is bounded above on compact S, while $\chi(q)$ is bounded below. Hence

$$v \leq \chi(q) + C_v$$

for a constant C_v. This is precisely the comparison expressing that T is no more singular than T'. Here global potentials exist by the $\partial\bar{\partial}$-lemma on the projective surface.

For this modified Grauert example, Proposition 2.4 therefore gives an affirmative answer to the bounded-potential question in [3, Section 4] for the modified example. That question also mentions Grauert's original example; the present argument does not identify its line bundle with L and does not settle that other case. The metric is constructed here, not taken from that preprint. This explicit answer for L is not a statement about arbitrary nef and big line bundles, nor a claim that every minimal-singularity representative has continuous potentials.

3. Line-bundle exhaustions and the Kähler counterexample

3.1. Dual norms and compact-group averaging.

Lemma 3.1. *Let Z be a compact complex manifold and $A \to Z$ a holomorphic line bundle with a positive continuous Hermitian metric h having psh local weights. Then the squared dual norm*

$$\rho(v) = \|v\|^2_{h^{-1}} \qquad (v \in \mathrm{Tot}(A^{-1}))$$

is a continuous psh exhaustion. If the weights are locally $C^{0,\gamma}$ for $0 < \gamma \leq 1$, so is ρ.

Proof. For a local frame e_i of A, write a dual vector as $v = \lambda_i e_i^*$, and write $h(e_i, e_i) = e^{-\varphi_i}$. Then

$$\rho = |\lambda_i|^2 e^{\varphi_i}. \tag{3.1}$$

Off the zero section, $\log\rho = \log|\lambda_i|^2 + \varphi_i$ is psh. Composition with the increasing convex function exp makes ρ psh there. Formula (3.1) extends continuously by zero to the zero section and is locally bounded, so removability gives psh on the whole total space.

Fix a smooth Hermitian metric h_{ref} on A^{-1}. By compactness of Z and positivity of h, constants $c, C > 0$ satisfy

$$c\|v\|^2_{h_{\mathrm{ref}}} \leq \rho(v) \leq C\|v\|^2_{h_{\mathrm{ref}}}.$$

The closed disk bundle of any fixed radius over Z is compact. Every sublevel of ρ is a closed subset of such a disk bundle and is therefore compact. The Hölder assertion follows locally from (3.1). □

Lemma 3.2 (Averaging preserves exhaustions)**.** *Let a compact group G act continuously on a locally compact Hausdorff space X, and let F be a continuous exhaustion. Its normalized Haar average*

$$\overline{F}(x) = \int_G F(gx)\, dg$$

is also an exhaustion. If G is a compact Lie group, X is a complex manifold, the action is smooth and holomorphic, and F is C^2 psh, then $\overline{F}$ is C^2 psh. The same assertion holds in the smooth category.

Proof. For $K_c = \{F \le c\}$, the orbit saturation $G\cdot K_c$ is compact. If $x \notin G\cdot K_c$, then $F(gx) > c$ for every $g \in G$, and continuity on the compact orbit implies $\overline{F}(x) > c$. Hence

$$\{\overline{F} \le c\} \subset G \cdot K_c.$$

The sublevel on the left is closed and therefore compact. Differentiation under the integral preserves C^2 regularity, and averaging the nonnegative Levi forms of the holomorphic pullbacks preserves psh. □

3.2. **The radius criterion.**

Proposition 3.3 (The radius criterion)**.** *Let Z be a compact complex manifold and $A \to Z$ a holomorphic line bundle. If $\mathrm{Tot}(A^{-1})$ admits a C^2 psh exhaustion, then A admits a C^2 Hermitian metric with semipositive curvature. If the exhaustion is smooth, the metric is smooth.*

Proof. Average the exhaustion over the holomorphic circle action $v \mapsto e^{i\alpha}v$ on the fibers, using Lemma 3.2. Denote the resulting invariant exhaustion by F. For a local frame e_i of A and $v = \lambda_i e_i^*$, circle invariance gives

$$F(z, \lambda_i) = U_i(z, t), \qquad t = \log|\lambda_i|^2, \quad \lambda_i \neq 0. \tag{3.2}$$

The function

$$V_i(z, \zeta) = U_i(z, \mathrm{Re}\,\zeta) = F(z, e^{\zeta/2}) \tag{3.3}$$

is a holomorphic pullback of F and is psh. In particular $U_{i,tt} \ge 0$.

For fixed z, continuity at the zero section gives the finite limit

$$\lim_{t\to-\infty} U_i(z, t) = F(z, 0).$$

Exhaustiveness gives $U_i(z, t) \to +\infty$ as $t \to +\infty$. A differentiable convex function with finite left limit has nonnegative derivative. Moreover, if its derivative vanishes at t_0, convexity forces it to be constant on $(-\infty, t_0]$. Consequently

$$U_{i,t}(z, t) > 0 \quad \text{whenever } U_i(z, t) > F(z, 0). \tag{3.4}$$

Fix $c > \max_Z F|_Z$. By convexity, the two limits, and (3.4), there is a unique solution

$$U_i(z, \sigma_i(z)) = c, \qquad U_{i,t}(z, \sigma_i(z)) > 0. \tag{3.5}$$

The implicit function theorem makes σ_i of class C^2, or smooth when F is smooth.

We verify the sign of its Levi form explicitly. Differentiating (3.5) twice in base coordinates gives

$$\begin{aligned} 0 = {} & U_{i,z_j\bar{z}_k} + U_{i,z_jt}\sigma_{i,\bar{z}_k} + U_{i,t\bar{z}_k}\sigma_{i,z_j} \\ & + U_{i,tt}\sigma_{i,z_j}\sigma_{i,\bar{z}_k} + U_{i,t}\sigma_{i,z_j\bar{z}_k}, \end{aligned} \tag{3.6}$$

with all derivatives of U_i evaluated at $(z, \sigma_i(z))$. For V_i in (3.3), its $\zeta\bar{\zeta}$ entry is $U_{i,tt}/4$, and its mixed entries have a factor $1/2$. Thus (3.6) is precisely the identity

$$\mathcal{L}_{V_i}(\xi, 2\partial\sigma_i(\xi)) = -U_{i,t}\mathcal{L}_{\sigma_i}(\xi). \tag{3.7}$$

The left side is nonnegative and $U_{i,t} > 0$, so $\psi_i = -\sigma_i$ is psh.

If $e_j = g_{ij}e_i$, the dual coordinates satisfy $\lambda_j = g_{ij}\lambda_i$. The level in (3.5) consequently transforms by

$$\sigma_j = \sigma_i + \log|g_{ij}|^2,$$

so

$$\psi_j = \psi_i - \log|g_{ij}|^2.$$

These are exactly the metric transition laws (1.3). Hence $\|e_i\|^2 = e^{-\psi_i}$ defines the desired semipositive Hermitian metric on A. □

Corollary 3.4. *For a holomorphic line bundle A over a compact complex manifold, $\mathrm{Tot}(A^{-1})$ admits a C^2 psh exhaustion if and only if A admits a C^2 semipositive Hermitian metric. The same equivalence holds with C^2 replaced by C^∞.*

Proof. The forward implication is Proposition 3.3. For the reverse implication, the squared dual norm in Lemma 3.1 has the regularity of the given metric, by its local expression (3.1). □

3.3. **Proof of the Kähler counterexample.**

Proof of Theorem 1.1. The construction in Subsection 2.1 gives a smooth projective surface S and an algebraic line bundle L. The total space M is the open subset

$$M = \mathbf{P}_{\mathrm{lines}}(\mathcal{O}_S \oplus L^{-1}) \setminus \mathbf{P}_{\mathrm{lines}}(L^{-1}). \tag{3.8}$$

The identification sends $v \in L_z^{-1}$ to the line $[(1, v)]$. The projective bundle in (3.8) is a smooth projective threefold. Restricting any of its Kähler forms proves that M is Kähler.

Let h be the continuous semipositive metric from Proposition 2.4. If $p : M \to S$ is the bundle projection, the exhaustion is explicitly

$$\rho(v) = \|v\|_{h^{-1}}^2 = \|v\|_{h_0^{-1}}^2 \exp\bigl(\chi(q(p(v)))\bigr). \tag{3.9}$$

Lemma 3.1 shows that ρ is continuous, psh, and exhaustive. Taking $a = \frac{1}{2}$ and the coefficient Λ supplied by Lemma 2.2 makes h, its weights, and ρ

locally Lipschitz. In particular the same example has local $C^{0,\gamma}$ regularity for all $0 < \gamma \leq 1$.

If a C^2 psh exhaustion on M existed, Proposition 3.3 would produce a C^2 semipositive Hermitian metric on L. Proposition 2.1 excludes such a metric. Therefore no C^2 psh exhaustion exists, completing the proof. □

Remark 3.5 (Interpretation). The example separates existence of a continuous exhaustion from existence of any smooth exhaustion. The conclusion does not depend on whether the particular function (3.9) can be smoothed: every hypothetical C^2 exhaustion leads to the forbidden metric. The argument does not determine whether a C^1 psh exhaustion exists, and the surface results below do not establish that dimension three is minimal. Thus a definition of weak pseudoconvexity requiring a smooth psh exhaustion is strictly stronger than the corresponding definition requiring only a continuous psh exhaustion, even for smooth quasi-projective Kähler threefolds.

4. Compact modifications and non-Kähler counterexamples

The Kähler counterexample can be changed over a single point to obtain a non-Kähler counterexample of the same dimension. We first isolate the exhaustion argument, then insert Hironaka's opposite-order blowup configuration in an exceptional plane. Compact products give a simpler alternative when an increase in dimension is allowed.

4.1. **Exhaustions under compactly supported modifications.**

Proposition 4.1. *Let $f : X' \to X$ be a proper surjective holomorphic map between nonempty complex manifolds. Suppose that, for a compact set $K \subset X$, it restricts to a biholomorphism*

$$X' \setminus f^{-1}(K) \longrightarrow X \setminus K.$$

For every $r \in \{0, 2, \infty\}$, X admits a C^r psh exhaustion if and only if X' does.

Proof. The case $K = \varnothing$ is immediate. Holomorphic pullback preserves plurisubharmonicity and the stated regularity, and properness makes the sublevels of a pulled-back exhaustion compact. Conversely, suppose F is a C^r psh exhaustion on X'. Choose a relatively compact open neighborhood U of K and

$$c > \max_{f^{-1}(\overline{U})} F.$$

Choose a smooth convex nondecreasing function $\chi : \mathbb{R} \to [0, \infty)$ that vanishes on $(-\infty, c]$ and tends to infinity. For example, integrate a smooth nondecreasing function which is zero up to c and one from $c + 1$ onward. On $X \setminus K$ define

$$G = \chi \circ F \circ (f|_{X' \setminus f^{-1}(K)})^{-1}.$$

This function is C^r and psh. It vanishes on $U \setminus K$, so extension by zero on K gives a C^r psh function on all of X. For each nonempty sublevel $\{G \le a\}$ choose b such that $\chi(t) \le a$ implies $t \le b$. Then

$$\{G \le a\} \subset K \cup f(\{F \le b\}).$$

The left side is closed and the right side is compact. Thus G is an exhaustion; no smooth descent of the original function F is needed. □

4.2. **A non-Kähler modification over one point.**

Proposition 4.2. *Let X be a connected smooth complex threefold and $x \in X$. There is a connected smooth non-Kähler threefold N and a proper holomorphic modification $\mu : N \to X$ inducing a biholomorphism*

$$N \setminus \mu^{-1}(x) \simeq X \setminus \{x\}.$$

Proof. Blow up x, obtaining

$$b : W = \mathrm{Bl}_x X \longrightarrow X, \qquad E = b^{-1}(x) \simeq \mathbf{P}^2.$$

In homogeneous coordinates $[u : v : w]$ on E, take

$$A = \{w = 0\}, \qquad B = \{uv = w^2\}.$$

The line A and smooth conic B meet transversely at exactly $p = [1 : 0 : 0]$ and $q = [0 : 1 : 0]$. Near either point they are the two coordinate axes in a smooth surface E inside W.

Over $W \setminus \{q\}$, first blow up $A \setminus \{q\}$ and then the strict transform of $B \setminus \{q\}$. Over $W \setminus \{p\}$, first blow up $B \setminus \{p\}$ and then the strict transform of $A \setminus \{p\}$. All these centers are closed smooth curves in the respective open manifolds. On $W \setminus \{p, q\}$ the centers are disjoint, so the blowups commute canonically over the base. Glue the two resulting manifolds through this isomorphism and write

$$\nu : N \longrightarrow W, \qquad \mu = b \circ \nu.$$

The glued space is Hausdorff: points with distinct images can be separated in W, and points with the same image belong to one of the Hausdorff open manifolds lying over the two base opens. It is smooth of dimension three because every blowup center is smooth. Properness is local on the target, so ν and μ are proper. Each blowup chart is connected and the charts overlap, hence N is connected. All centers lie in E, so μ is an isomorphism away from x. This is Hironaka's opposite-order construction [20, Section 6]; see also [21, Tag 08J1].

For completeness, the obstruction to a Kähler form is entirely contained in compact exceptional surfaces. Let $D_A, D_B \subset N$ be the exceptional divisors dominating A, B, and let $\alpha, \beta \in H_2(N, \mathbb{R})$ be the classes of their general $\mathbf{P}^1$ fibers. The surfaces D_A, D_B are smooth and compact: locally each is a ruled surface or its blowup at a point, and they map properly to the compact curves A, B. The fiber divisors of each map are linearly equivalent, since $A \simeq B \simeq \mathbf{P}^1$, and hence are homologous on the corresponding compact surface.

Over p, the strict transform of B meets the first exceptional divisor transversely at one point. The second blowup therefore restricts on that divisor to a surface blowup at that point. Its fiber over p is the reduced union of the strict transform $R_p \simeq \mathbf{P}^1$ of the first exceptional fiber and the new exceptional fiber $T_p \simeq \mathbf{P}^1$. Both occur with multiplicity one. Indeed, take local coordinates (x, y, z) with $A = \{y = z = 0\}$ and $B = \{x = z = 0\}$. In the first blowup chart $z = yt$, the first exceptional divisor is $y = 0$ and the second center is $x = t = 0$. Its restriction to $y = 0$ is the blowup of the smooth surface with coordinates (x, t) at the origin. The pullback of the fiber equation $x = 0$ is its strict transform plus the exceptional curve, each once. The curve T_p is also the fiber of $D_B \to B$ over p. Thus

$$\alpha = [R_p] + [T_p], \qquad \beta = [T_p].$$

Over q the order is reversed. With analogous notation,

$$\beta = [R_q] + [T_q], \qquad \alpha = [T_q].$$

These relations hold in ordinary homology: they arise from fiber divisors on compact D_A and D_B, regardless of compactness of N. Adding them gives

$$[R_p] + [R_q] = 0 \quad \text{in } H_2(N, \mathbb{R}). \tag{4.1}$$

If ω were a Kähler form on N, closedness and (4.1) would imply

$$0 = \int_{R_p} \omega + \int_{R_q} \omega > 0,$$

a contradiction. □

Proof of Theorem 1.2. Apply Proposition 4.2 to the threefold M of Theorem 1.1 at the chosen point x. If ρ is its continuous psh exhaustion, then

$$\rho_N = \rho \circ \mu, \qquad \{\rho_N \le a\} = \mu^{-1}(\{\rho \le a\})$$

is a continuous psh exhaustion of N. A holomorphic map is locally Lipschitz in smooth coordinate charts, so local Lipschitz regularity of the exhaustion is preserved. A C^2 psh exhaustion on N would give one on M by Proposition 4.1 with $K = \{x\}$, contrary to Theorem 1.1. □

Remark 4.3 (The sense in which the change is local). The point x can be chosen outside the zero section of $M \to S$. The construction then preserves a neighborhood of that section and, by properness, a neighborhood of infinity. More precisely, for every neighborhood U of x, μ is an isomorphism over $M \setminus U$. This is a modification supported over one point; it makes no claim about a small deformation of the complex structure on a fixed smooth manifold.

4.3. **Compact products and an explicit fivefold.**

Proposition 4.4 (Compact product principle)**.** *Let X be a nonempty complex manifold and T a nonempty compact connected complex manifold. For each $r \in \{0, 2, \infty\}$, X admits a C^r psh exhaustion if and only if $X \times T$ does. If T is non-Kähler, then $X \times T$ is non-Kähler.*

Proof. Pullback by the projection preserves regularity and plurisubharmonicity, with compact sublevels

$$\{f \circ \mathrm{pr}_X \leq a\} = \{f \leq a\} \times T. \tag{4.2}$$

Conversely, restriction to the closed holomorphic slice $X \times \{t_0\}$ preserves regularity, plurisubharmonicity, and compact sublevels. A Kähler form on the product would restrict to one on $\{x_0\} \times T$, proving the final assertion. In fact, the maximum principle makes every continuous psh function on the product constant on each compact connected fiber $\{x\} \times T$. □

Corollary 4.5 (A non-Kähler fivefold)**.** *Let M be the threefold of Theorem 1.1 and let*

$$H = (\mathbb{C}^2 \setminus \{0\})/\langle z \mapsto 2z \rangle. \tag{4.3}$$

Then $M \times H$ is a smooth non-Kähler fivefold admitting a continuous psh exhaustion but no C^2 psh exhaustion. For every $0 < \gamma < 1$ its exhaustion can be chosen locally $C^{0,\gamma}$.

Proof. The dilation action is free and properly discontinuous: if a compact set lies in $r \leq \|z\| \leq R$, only finitely many integers n satisfy $r/R \leq 2^n \leq R/r$. Polar coordinates identify its quotient smoothly with

$$H \simeq S^3 \times (\mathbb{R}/(\log 2)\mathbb{Z}) \simeq S^3 \times S^1. \tag{4.4}$$

Thus H is a compact connected complex surface with $H^2_{\mathrm{dR}}(H) = 0$. Any Kähler form ω would be exact, and Stokes' theorem would give $\int_H \omega^2 = 0$, contradicting positivity. Apply Proposition 4.4 and Theorem 1.1. Explicitly,

$$\rho_{M \times H}(v, [z]) = \rho(v) \tag{4.5}$$

is the required exhaustion, with the same local Hölder regularity. □

5. The surface problem and boundary reductions

5.1. **Line-bundle total spaces over compact curves.** The construction of Theorem 1.1 has a two-dimensional compact base and a one-dimensional fiber. Its dimension cannot be lowered by replacing the base by a compact Riemann surface: on a compact curve, pseudoeffectivity of a line bundle already implies smooth semipositivity. Here pseudoeffectivity means the existence of a closed positive $(1,1)$-current in the first Chern class; its potentials may be singular. Thus this assertion is stronger than the analogous assertion for a continuous semipositive metric.

Proposition 5.1 (Semipositivity on a compact curve)**.** *Let B be a compact connected Riemann surface and A a holomorphic line bundle on B. The following conditions are equivalent:*

(i) *A is pseudoeffective;*
(ii) $\deg A \geq 0$*;*
(iii) *A admits a smooth Hermitian metric with semipositive curvature.*

For $\deg A = 0$ *the metric can be chosen flat, and for* $\deg A > 0$ *it can be chosen to have strictly positive curvature.*

Proof. A positive $(1,1)$-current on a curve is a positive measure. If it represents $c_1(A)$, its total mass is $\deg A$, proving (i)⇒(ii). For the converse construction, choose a smooth Hermitian metric h_0 on A, with curvature θ_0, and a smooth positive $(1,1)$-form ω with $\int_B \omega = 1$. Write $d = \deg A$. Since

$$\int_B (d\omega - \theta_0) = 0,$$

the Poisson equation, equivalently the $\partial\bar\partial$-lemma on a compact Riemann surface, gives a smooth real function f with

$$dd^c f = d\omega - \theta_0.$$

The metric $h = h_0 e^{-f}$ has curvature

$$\Theta_h(A) = \theta_0 + dd^c f = d\omega.$$

It is semipositive if $d \geq 0$, flat if $d = 0$, and strictly positive if $d > 0$. Its curvature itself is a positive current, completing the equivalence. □

Remark 5.2. This is the classical degree criterion, and does not assert semiampleness. A non-torsion line bundle in $\mathrm{Pic}^0(B)$ is unitary flat, but has no nonzero section of any positive tensor power. For a noncompact Riemann surface, every holomorphic line bundle is holomorphically trivial: the surface is Stein, and the exponential sequence gives this assertion from $H^1(B, \mathcal{O}_B) = 0$ and $H^2(B, \mathbb{Z}) = 0$. The compact case is the one relevant to the construction in this paper.

Corollary 5.3 (Line-bundle total spaces of dimension two)**.** *For a holomorphic line bundle* A *on a compact connected Riemann surface* B*, the following are equivalent:*

(i) $\mathrm{Tot}(A^{-1})$ *admits a continuous psh exhaustion;*
(ii) $\deg A \geq 0$*;*
(iii) $\mathrm{Tot}(A^{-1})$ *admits a smooth psh exhaustion.*

Proof. If $\deg A \geq 0$, Proposition 5.1 and Lemma 3.1 give the smooth squared dual norm exhaustion. For the remaining implication, compactify the total space as

$$\mathrm{Tot}(A^{-1}) = \mathbf{P}_{\mathrm{lines}}(\mathcal{O}_B \oplus A^{-1}) \setminus C_\infty, \qquad C_\infty = \mathbf{P}_{\mathrm{lines}}(A^{-1}).$$

Its boundary normal bundle is

$$N_{C_\infty/\mathbf{P}(\mathcal{O}_B \oplus A^{-1})} \simeq \mathrm{Hom}(A^{-1}, \mathcal{O}_B) \simeq A.$$

If $\deg A < 0$, this boundary is negative definite, and Proposition 5.10 below rules out a continuous psh exhaustion of its complement. □

In particular, a continuous semipositive metric on a line bundle over a compact curve cannot furnish the metric obstruction used in Theorem 1.1: a smooth semipositive metric always exists on that same bundle. This rules out the present construction in complex dimension two; it does not prove the equivalence of the two exhaustion definitions for arbitrary complex surfaces.

5.2. **Historical surface results and their precise scope.** The surface problem has two different aspects: regularizing an exhaustion, and deciding when the surface is holomorphically convex. The latter property gives a regularization mechanism through a proper holomorphic map, but it is stronger than the existence of a smooth psh exhaustion.

Ohsawa's 1981 result [4, Proposition 1.4] states that a connected noncompact smooth weakly 1-complete surface is holomorphically convex if and only if it has a nonconstant holomorphic function. His definition of weak 1-completeness assumes a C^∞ psh exhaustion. The remark following his Theorem 1.1 says that continuity suffices for that theorem's dichotomy concerning the fibers of a holomorphic function; it should not be read as changing the regularity hypothesis throughout the paper.

Tomassini and Vâjâitu subsequently treated singular surfaces with a continuous psh exhaustion [5]; see also the more general holomorphic-convexity theorem [6, Theorem 1.1]. We use only its consequence for a smooth connected surface equipped with a finite continuous psh exhaustion. The following proposition records that consequence, rather than a new holomorphic-convexity theorem. No assertion about regularizing an arbitrary possibly discontinuous exhaustion is needed.

Proposition 5.4 (Consequence of Vâjâitu)**.** *Let X be a connected noncompact complex surface admitting a continuous psh exhaustion. If $\mathcal{O}(X) \neq \mathbb{C}$, then X admits a smooth psh exhaustion.*

Proof. Vâjâitu's theorem makes X holomorphically convex. Its Remmert reduction is a proper holomorphic map $r : X \to Y$ to a Stein space. Choose a smooth strongly psh exhaustion η on Y, where smoothness on a complex space means local restriction of a smooth function from a local ambient Euclidean space. Then $\eta \circ r$ is smooth on X, including along the positive-dimensional fibers, and it is psh by holomorphic pullback. Finally,

$$\{\eta \circ r \leq t\} = r^{-1}(\{\eta \leq t\})$$

is compact by properness. Thus $\eta \circ r$ is an exhaustion. Its Levi form may degenerate along compact fibers, as is appropriate for weak 1-completeness. □

This implication does not dispose of surfaces for which $\mathcal{O}(X) = \mathbb{C}$. In particular, a holomorphically convex noncompact connected surface always has a nonconstant holomorphic function, whereas a smooth psh exhaustion can exist without one. Thus replacing the holomorphic-function hypothesis in this argument requires another regularization mechanism.

For a continuously weakly complete manifold X, its continuous minimal kernel is

$$\Sigma_X^0 = \bigcap_v \{x \in X : v \text{ is not strongly psh near } x\},$$

where v ranges over continuous psh exhaustions. Strong plurisubharmonicity here means stability under sufficiently small C^2 perturbations locally.

Proposition 5.5 (Mongodi–Slodkowski criterion)**.** *Let X be a connected complex manifold with a continuous psh exhaustion u. If $u(\Sigma_X^0)$ contains no terminal half-line $(a, +\infty)$, then X admits a smooth psh exhaustion. Consequently, if a continuously weakly complete manifold has no smooth psh exhaustion, then for every continuous psh exhaustion u there is $a_u \in \mathbb{R}$ such that*

$$(a_u, +\infty) \subset u(\Sigma_X^0).$$

Proof. By [7, Lemma 2.1], the stated image condition makes X a modification of a Stein space. Pulling back a smooth strongly psh exhaustion by this proper holomorphic map gives the claim. The final assertion is the contrapositive. No ambient real-analytic exhaustion is assumed in this lemma. □

A separate theorem in [7, Corollaries 4.4–4.5] requires a noncompact surface M already carrying a real-analytic psh exhaustion. Every continuously weakly complete domain $\Omega \subset M$ then has a real-analytic psh exhaustion, strongly psh outside Σ_Ω^0, and its continuous and real-analytic minimal kernels coincide. Their Theorem 1.1 also describes the domain's geometry through Stein modifications, proper fibrations, or Grauert-type structure.

The ambient hypothesis needs independent verification: taking $M = \Omega$ is circular, and a compactification $\Omega = S \setminus |D|$ does not supply the required noncompact ambient surface. These results do not prove the unrestricted implication or settle every projective divisor complement. This limitation does not assert that the unrestricted problem has any particular published status.

5.3. **Divisorial compactifications and allowed modifications.** For the divisor-complement arguments in the rest of this section, S is a connected smooth compact complex surface, D is a nonzero effective divisor, and

$$X = S \setminus \operatorname{supp} D.$$

The existence of this compactification is an additional hypothesis on X, rather than a property of every noncompact complex surface. Only the support of D determines the complement. We consequently replace D by its reduced divisor whenever convenient. Kählerness or projectivity will be imposed explicitly when needed.

Lemma 5.6. *If u is a continuous exhaustion of X, then $u(x) \to +\infty$ as $x \to \operatorname{supp} D$. More precisely, for every $b \in \mathbb{R}$ there is a neighborhood U_b of $\operatorname{supp} D$ in S such that $u > b$ on $U_b \setminus \operatorname{supp} D$.*

Proof. The sublevel set $K_b = \{u \le b\}$ is a compact subset of X, hence a closed subset of S disjoint from $\operatorname{supp} D$. Take $U_b = S \setminus K_b$. □

Proposition 5.7. *Let $\mu : X' \to X$ be a proper modification between connected smooth complex surfaces with compact exceptional locus. For each $\mathcal{R} \in \{C^0, C^2, C^\infty\}$, X admits a psh exhaustion of class $\mathcal{R}$ if and only if X' does.*

Proof. The image of the exceptional locus is compact, and μ is an isomorphism outside that image. Apply Proposition 4.1. □

Remark 5.8. For a blowup $\mu : S' \to S$ at a boundary point, the appropriate upstairs boundary is the full reduced inverse image of $\operatorname{supp} D$, including the exceptional curve. Its complement is biholomorphic to X. In particular, resolving the boundary to simple normal crossings does not change the open surface. Keeping only the strict transform of the boundary generally changes the complement and does not give a proper modification of the original open surface. Proposition 5.7 concerns modifications with compact exceptional locus; it does not assert descent under arbitrary modifications with infinitely many exceptional centers escaping to infinity.

5.4. **An obstruction from negative definite boundary components.**

Lemma 5.9. *Let (V, p) be a normal complex surface germ with an isolated puncture. A continuous plurisubharmonic function on $V \setminus \{p\}$ is bounded above on a sufficiently small punctured neighborhood of p.*

Proof. A local finite projection gives a proper finite holomorphic map $\tau : V \to \Delta^2$ after shrinking representatives, with $\tau^{-1}(0) = \{p\}$. If v is the given function, put

$$v_*(z) = \max\{v(y) : \tau(y) = z\}, \qquad z \neq 0.$$

This function is continuous, by properness and openness of the finite map. Away from its branch locus it is the maximum of finitely many plurisubharmonic functions on local inverse branches. Along the branch locus away from 0 it is locally bounded and continuous, so the usual removable-singularity theorem makes it plurisubharmonic there as well.

For completeness, a Hartogs argument gives the needed bound on $\Delta^2 \setminus \{0\}$. Choose a small bidisc and $r > 0$ so that its compact vertical circular boundary

$$K = \{(z_1, z_2) : |z_1| \le r,\ |z_2| = r\}$$

lies in the punctured bidisc. Set $M = \max_K v_*$. For $z_1 \neq 0$, the maximum principle on the vertical disc gives $v_*(z_1, z_2) \le M$ for $|z_2| < r$. For $z_1 = 0$ and $z_2 \neq 0$, the same bound follows by continuity. Since $v \le v_* \circ \tau$, this bounds v near p. □

Proposition 5.10. *Suppose that a connected component E of $\operatorname{supp} D$ has negative definite intersection matrix. Then $S \setminus \operatorname{supp} D$ admits no continuous plurisubharmonic exhaustion.*

Proof. Grauert's contraction theorem [8] contracts E, on a neighborhood U avoiding the other boundary components, to an isolated point p of a normal complex surface V. The contraction induces a biholomorphism $U\backslash E \simeq V\backslash\{p\}$. An exhaustion u on X would induce a continuous plurisubharmonic function on this punctured neighborhood. Lemma 5.9 makes it bounded above near p, whereas Lemma 5.6 requires it to tend to infinity. □

5.5. **Intersection matrices and smooth collars.**

Lemma 5.11. *Suppose that each connected component E of* $\operatorname{supp} D$ *has a neighborhood U_E and a smooth plurisubharmonic function $\Phi_E : U_E \setminus E \to \mathbb{R}$ tending to infinity at E. Then X admits a smooth plurisubharmonic exhaustion.*

Proof. There are finitely many boundary components. Choose pairwise disjoint smaller neighborhoods W_E whose closures lie in U_E, with compact boundaries disjoint from E. Choose $T_E > \max_{\partial W_E} \Phi_E$. Let χ_E be smooth, convex, nondecreasing, zero on $(-\infty, T_E]$, and unbounded at infinity. Continuity ensures that $\chi_E(\Phi_E)$ vanishes on a neighborhood of ∂W_E. Therefore the function equal to $\chi_E(\Phi_E)$ on $W_E \setminus E$ and zero elsewhere on X is smooth and plurisubharmonic. The sum of these functions tends to infinity at every boundary component. Since S is compact, its sublevel sets are compact subsets of X. □

Lemma 5.12. *Let $E = \bigcup_{i=1}^{r} E_i$ be a connected reduced simple normal crossing divisor on a smooth compact complex surface. If a divisor $A = \sum_i a_i E_i$, with $a_i \in \mathbb{Z}_{>0}$, satisfies $A \cdot E_i > 0$ for every i, then $\mathcal{O}_S(A)$ has a smooth Hermitian metric of strictly positive curvature on a neighborhood of E.*

Proof. The restriction $\mathcal{O}_S(A)|_E$ is ample, since its degree on every irreducible component is positive. Choose $m > 0$ for which $\mathcal{O}_S(mA)|_E$ is very ample, and endow this restriction with its Fubini–Study metric. On a sufficiently small ambient coordinate neighborhood, holomorphic sections giving the embedding of E lift to holomorphic local sections of $\mathcal{O}_S(mA)$. Their Fubini–Study weights, divided by m, define local smooth metrics of $\mathcal{O}_S(A)$, with the prescribed restriction to E and positive curvature in tangent directions along its smooth part.

At a node the Zariski tangent space of E equals the ambient tangent space. A projective embedding of E is injective on that tangent space. Accordingly, the local Fubini–Study metric obtained by lifting its sections has strictly positive ambient curvature at the node, and on a smaller neighborhood of it. Fix these local metrics on disjoint small neighborhoods of the finitely many nodes. Patch the remaining local extensions of the prescribed metric using a smooth partition of unity, relative to any global reference metric, while keeping the chosen metrics unchanged on smaller node neighborhoods. On the smooth part of E, the tangential curvature is unchanged by this patching: tangential second derivatives depend only on the common restriction to

the curve. The resulting global smooth metric thus has positive tangential curvature on E and strictly positive ambient curvature near its nodes.

Let s_E be the canonical section of $\mathcal{O}_S(E)$, choose any smooth metric on this bundle, and set $q = \|s_E\|^2$. Along the smooth part of E, $dd^c q$ vanishes on the tangent line and is strictly positive in the normal direction. At a node $dd^c q = 0$, because a reduced local equation is the product of the two branch coordinates. In particular, $dd^c q$ is positive semidefinite along all of E.

Add Cq to the local weights of the patched metric. Away from fixed node neighborhoods, compactness, positivity of the tangential block, and positivity of the normal block of $dd^c q$ show that a sufficiently large C makes the full curvature matrix positive along E; this is also immediate from its Schur complement. Near the nodes, the old curvature is already positive and $dd^c q$ is semipositive on E. The new curvature is therefore positive everywhere on E, hence on an ambient neighborhood by continuity. □

Proposition 5.13. *Assume that S is Kähler, that D is reduced with simple normal crossings and that X admits a continuous plurisubharmonic exhaustion. For each connected component $E = \bigcup_{i=1}^{r} E_i$ of D, let $I = (E_i \cdot E_j)$ and let λ be its largest eigenvalue. Then $\lambda \geq 0$, and the following alternatives hold.*

(i) *If $\lambda > 0$, a smooth strictly plurisubharmonic collar exhaustion exists near E.*

(ii) *If $\lambda = 0$, there are positive integers a_i such that $A = \sum_i a_i E_i$ is nef and*

$$A \cdot E_i = 0 \quad \text{for every } i, \qquad A^2 = 0.$$

The matrix I is negative semidefinite with one-dimensional kernel.

Thus the only connected boundary configurations not already settled by the collar construction are supports of effective nef square-zero divisors as in (ii).

Proof. The off-diagonal entries of I are nonnegative, and connectedness of E makes the matrix irreducible. Applying Perron–Frobenius to $I + c\,\mathrm{Id}$ for sufficiently large c, its largest eigenvalue is simple and has a strictly positive eigenvector. If $\lambda < 0$, I is negative definite, contrary to Proposition 5.10.

If $\lambda > 0$, approximate its positive eigenvector by a positive rational vector a sufficiently closely that Ia has strictly positive entries, and clear denominators. Then $A \cdot E_i > 0$ for all i. Lemma 5.12 gives a positive metric h on a neighborhood of E. The function

$$\Phi = -\log \|s_A\|_h^2$$

is smooth and strictly plurisubharmonic off E, since $dd^c\Phi = \Theta_h(\mathcal{O}_S(A))$ there. It tends to infinity at every point of E.

If $\lambda = 0$, every eigenvalue is nonpositive, and simplicity of the largest eigenvalue gives a one-dimensional kernel spanned by a positive vector. Since I has integral entries, its kernel has a rational generator; clear denominators to obtain positive integers a_i. Then $A \cdot E_i = 0$ and $A^2 = 0$. For any irreducible curve G outside E, effectiveness gives $A \cdot G \geq 0$. In the projective case this

proves nefness. For a nonprojective compactifier, the independent argument in Corollary 5.17 below identifies E as a whole elliptic fiber support. Its positive kernel vector is proportional to the fiber multiplicity vector, so some positive multiple of A is a positive multiple of a fiber divisor. Pulling back a positive metric from the base and taking its appropriate root gives a smooth semipositive metric on $\mathcal{O}_S(A)$, proving nefness also in the analytic sense. Its numerical class is nonzero, because $A \cdot H > 0$ for a Kähler class H. □

5.6. **Disconnected boundaries and nonprojective compactifiers.**

Theorem 5.14. *Let S be smooth projective and let D be a nonzero effective divisor whose support is disconnected. If $S \setminus \operatorname{supp} D$ admits a continuous plurisubharmonic exhaustion, then it admits a smooth one.*

Proof. Resolve the boundary using its full reduced inverse image, which preserves both the complement and the number of its connected boundary components. Write these components as $E_1, \dots, E_r$, with $r \geq 2$. Proposition 5.13 supplies an effective divisor A_k supported exactly on E_k, with $A_k^2 \geq 0$; in the positive-eigenvalue case its square is strictly positive. Disjointness gives $A_k \cdot A_l = 0$ for $k \neq l$.

If $A_k^2 > 0$, the Hodge index theorem makes the intersection form negative definite on its orthogonal complement. The numerical class of A_l is nonzero, because its intersection with an ample divisor is positive. We would therefore have $A_l^2 < 0$, a contradiction. Thus every boundary component is in the zero-eigenvalue alternative:

$$A_k \text{ is nef}, \qquad A_k^2 = 0, \qquad A_k \cdot A_l = 0.$$

The Hodge index theorem also implies that these nonzero isotropic nef classes are proportional. Their proportionality factors are positive and rational, as can be seen by intersecting them with an integral ample divisor. For every $k \geq 2$, choose positive integers p_k, q_k such that $p_k A_k - q_k A_1$ is numerically trivial.

On a smooth projective surface, numerical triviality of a divisor implies that its real first Chern class vanishes. Equivalently, this follows from the Hodge index theorem on divisor classes. The $\partial\bar{\partial}$-lemma then supplies a smooth unitary flat Hermitian metric h_k on

$$L_k = \mathcal{O}_S(p_k A_k - q_k A_1).$$

Indeed, the curvature of any smooth reference metric is an exact real $(1,1)$-form, and changing its weight by a global smooth potential makes its curvature zero. This argument also permits torsion integral first Chern classes.

Let σ_k be the canonical meromorphic section of L_k with divisor $p_k A_k - q_k A_1$. On $X = S \setminus \operatorname{supp} D$, the function

$$\phi_k = \log \|\sigma_k\|_{h_k}^2$$

is smooth and pluriharmonic. It tends to $-\infty$ at E_k and to $+\infty$ at E_1. Hence

$$\rho = \sum_{k=2}^{r} \phi_k^2$$

is smooth and plurisubharmonic, because $dd^c(\phi_k^2) = \frac{i}{\pi}\partial\phi_k \wedge \bar{\partial}\phi_k \geq 0$. It tends to infinity at every boundary component. Compactness of S makes it an exhaustion. The same function transports through the biholomorphism of complements obtained from the boundary resolution. □

Theorem 5.15 (A numerical criterion at nonprojective infinity)**.** *Let S be a smooth compact connected nonprojective complex surface and $D \neq 0$ an effective divisor. Put $X = S \setminus \operatorname{supp} D$. The following are equivalent:*

(i) *X admits a continuous plurisubharmonic exhaustion;*
(ii) *X admits a smooth plurisubharmonic exhaustion;*
(iii) *no connected component of $\operatorname{supp} D$ has negative-definite intersection matrix.*

There is no Kähler assumption, and the boundary need not have normal crossings.

Proof. Proposition 5.10 proves (i)⇒(iii), without a Kähler hypothesis. The implication (ii)⇒(i) is immediate. We prove (iii)⇒(ii).

The algebraic dimension of S is zero or one. If $a(S) = 1$, its algebraic reduction $f : S \to B$ is a holomorphic elliptic fibration with connected fibers and every compact curve is vertical; see [9, Section 2]. For a fiber $F = \sum m_i C_i$, its intersection matrix I satisfies

$$x^{\mathsf{T}} I x = -\sum_{i<j} (C_i \cdot C_j) m_i m_j \left(\frac{x_i}{m_i} - \frac{x_j}{m_j} \right)^2 . \tag{5.1}$$

Connectedness makes its kernel the line spanned by (m_i). Every proper principal submatrix is consequently negative definite. Condition (iii) therefore forces the boundary to contain every fiber support it meets. Hence $X = f^{-1}(B \setminus P)$ for a finite nonempty set P, and the restriction of f is proper. Pull back a smooth strictly subharmonic exhaustion of $B \setminus P$.

Suppose $a(S) = 0$. By [9, Lemma 2.1(1)], $c_1(L)^2 \leq 0$ for every holomorphic line bundle L on S. Thus every boundary intersection matrix is negative semidefinite. On each connected component, condition (iii) and Perron–Frobenius give a strictly positive kernel vector; the integral matrix permits an integral choice after rescaling. Summing the corresponding divisors gives

$$A = \sum a_i D_i, \qquad a_i > 0, \qquad A^2 = 0, \qquad \operatorname{supp} A = \operatorname{supp} D.$$

Contract to a smooth minimal model $p : S \to S_0$. For any single blowdown $q : T \to T'$ with exceptional curve E, put $A' = q_* A$ and write $A = q^* A' + bE$. Then

$$0 = A^2 = (A')^2 - b^2.$$

The surface T' remains nonprojective, so $(A')^2 \leq 0$ by the same lemma. Consequently $b = 0$, $(A')^2 = 0$, and $A' \neq 0$. Iterating gives $A = p^*A_0$ for a nonzero effective square-zero divisor A_0 and, crucially,

$$\operatorname{supp} A = p^{-1}(\operatorname{supp} A_0).$$

Thus $X = p^{-1}(X_0)$ for $X_0 = S_0 \setminus \operatorname{supp} A_0$, and $p|_X$ is proper. A smooth psh exhaustion of X_0 pulls back to one of X.

If S is Kähler, its minimal model of algebraic dimension zero is a torus or a K3 surface [9, Section 2]. A torus of algebraic dimension zero has no curves: a curve on a nonprojective torus must be a smooth elliptic curve by adjunction and the absence of rational curves, and its elliptic quotient would give algebraic dimension at least one. On a K3 surface, Riemann–Roch and $H^0(S_0, \mathcal{O}(-A_0)) = 0$ yield $h^0(S_0, \mathcal{O}(A_0)) \geq 2$, contradicting $a(S_0) = 0$. Thus this alternative cannot occur.

If S is non-Kähler, S_0 is of class VII; see [9, Section 2]. For $b_2(S_0) = n > 0$, Enoki's Main Theorem [10, p. 454] identifies S_0 with $S_{n,\alpha,t}$ and A_0 with a positive multiple of its distinguished rational cycle. Its complement is an affine-line bundle of degree $-n$ over an elliptic curve [10, (0.4), p. 453]. Completing the affine fibers gives a smooth projective $\mathbf{P}^1$-bundle whose infinity section C_∞ has square $n > 0$. Lemma 5.12 gives a metric of positive curvature near C_∞. The function $-\log \|s_{C_\infty}\|^2$ is a smooth strictly psh divergent collar, and Lemma 5.11 gives a smooth exhaustion of X_0.

For $b_2(S_0) = 0$, a minimal class VII surface with a curve is elliptic or Hopf by the theorem of Kodaira recalled in [10, p. 453]. Algebraic dimension zero excludes the elliptic-fibration case, so S_0 is Hopf. Every irreducible curve C on S_0 has $C^2 = K_{S_0} \cdot C = 0$, hence arithmetic genus one. Its normalization cannot be rational: any holomorphic map from $\mathbf{P}^1$ lifts to the universal cover $\mathbb{C}^2 \setminus \{0\}$ and is constant. Therefore C is smooth elliptic. Distinct such curves are disjoint because their intersection numbers vanish. The boundary of X_0 is consequently a disjoint union of smooth square-zero elliptic curves. Finite Ueda type gives a smooth strictly psh collar by [14, Theorem 2.5]; infinite type gives a unitary flat neighborhood by [16, Theorem 1.6]. The latter gives a smooth pluriharmonic logarithmic collar. Lemma 5.11 combines these collars into a smooth psh exhaustion of X_0. □

Remark 5.16. The Hopf elliptic-neighborhood step uses the September 2026 preprint [16]. Enoki's theorem applies to any nonzero square-zero divisor on a minimal class VII surface with positive second Betti number; it does not assume that the divisor is already a cycle. Thus the proof does not use an unresolved classification of all class VII surfaces. It also explains why the whole pullback of the boundary, rather than its strict transform alone, matters when passing to the minimal model.

Corollary 5.17 (The nonprojective Kähler case)**.** *Let S be a connected smooth compact nonprojective Kähler surface and $D \neq 0$ effective. If $S \setminus \operatorname{supp} D$ has a continuous psh exhaustion, then $a(S) = 1$, and the algebraic*

reduction is a holomorphic elliptic fibration $f : S \to B$ with connected fibers such that

$$\operatorname{supp} D = f^{-1}(P)$$

for a finite nonempty subset $P \subset B$. A smooth exhaustion is pulled back from $B \setminus P$.

Proof. The proof of Theorem 5.15 excludes algebraic dimension zero in the Kähler case and proves the whole-fiber description in algebraic dimension one, including for nonminimal surfaces and multiple fibers. □

5.7. **Moving boundaries and the residual projective case.**

Proposition 5.18. *Let S be projective and let A be a nonzero effective semiample divisor with $A^2 = 0$. Then $S \setminus \operatorname{supp} A$ admits a smooth plurisubharmonic exhaustion. There are a smooth projective curve B, a proper holomorphic map $f : S \to B$ with connected fibers, a positive integer m, and a nonzero effective divisor G on B such that*

$$mA = f^*G.$$

Proof. A globally generated multiple of A defines a projective morphism. Its image is a curve: it is not a point because A is numerically nonzero, and it cannot have dimension two because $A^2 = 0$. Its Stein factorization gives $f : S \to B$ with B a smooth projective curve and connected fibers, and $\mathcal{O}_S(mA) = f^*L$ for a line bundle L on B after increasing m if necessary. Since $f_*\mathcal{O}_S = \mathcal{O}_B$, the canonical section with divisor mA is the pullback of a section of L. Its divisor G is effective and nonzero. Consequently,

$$S \setminus \operatorname{supp} A = f^{-1}(B \setminus \operatorname{supp} G),$$

and the restriction of f to this complement is proper.

Every compact Riemann surface with a nonempty finite set removed has a smooth strictly subharmonic exhaustion ψ. Explicitly, fix an area form ω of total mass one, and for each removed point p take a Green function g_p with $dd^c g_p = \delta_p - \omega$. Then $\psi = -\sum_p g_p$ is smooth with $dd^c\psi = \#\{p\}\,\omega > 0$ on the punctured curve and tends to infinity at each puncture. The pullback $\psi \circ f$ is the required smooth plurisubharmonic exhaustion. □

Proposition 5.19. *Let S be smooth projective and let A be a nonzero effective nef divisor with $A^2 = 0$. If either $\kappa(S, A) = 1$ or $K_S \cdot A < 0$, then $S \setminus \operatorname{supp} A$ admits a smooth plurisubharmonic exhaustion.*

Proof. A nef line bundle of Iitaka dimension one on a compact normal surface is semiample by [12, Lemma 7.3]. Thus the first case follows from Proposition 5.18.

For the second case, Riemann–Roch gives

$$\chi(\mathcal{O}_S(nA)) = \chi(\mathcal{O}_S) - \frac{n}{2} K_S \cdot A.$$

For an ample divisor H, the intersection $(K_S - nA) \cdot H$ is negative for large n, since $A \cdot H > 0$. Therefore $H^0(S, \mathcal{O}_S(K_S - nA)) = 0$ and, by Serre

duality, $H^2(S, \mathcal{O}_S(nA)) = 0$. It follows that $h^0(S, \mathcal{O}_S(nA)) \geq \chi(\mathcal{O}_S(nA))$ is unbounded when $K_S \cdot A < 0$. Consequently $\kappa(S, A) \geq 1$. A nef divisor of square zero has Iitaka dimension at most one: a moving linear system with two-dimensional image would have a positive-square moving part on a resolution, contradicting nefness and $A^2 = 0$. Thus $\kappa(S, A) = 1$, and the first case applies. □

Corollary 5.20. *For divisorial compactifications by smooth projective surfaces, the unproved continuous-to-smooth exhaustion implication reduces to the following class of pairs:*

$$D \text{ reduced with simple normal crossings and connected support,}$$

$$A = \sum_i a_i D_i, \quad a_i \in \mathbb{Z}_{>0}, \quad A \text{ nef,}$$

$$A \cdot D_i = 0 \text{ for every } i, \qquad A^2 = 0,$$

$$\kappa(S, A) = 0, \qquad K_S \cdot A \geq 0.$$

In this reduction, the standing hypothesis that $S \setminus \operatorname{supp} D$ has a continuous plurisubharmonic exhaustion is retained.

Proof. Boundary resolution preserves the open surface. Disconnected supports are handled by Theorem 5.14. Proposition 5.13 either gives a smooth collar, hence a global exhaustion, or supplies A with the displayed intersection properties. Since A is nonzero and effective, $\kappa(S, A) \geq 0$; nefness and $A^2 = 0$ give $\kappa(S, A) \leq 1$. Proposition 5.19 settles $\kappa(S, A) = 1$ and $K_S \cdot A < 0$. The stated cases remain. □

6. Birational reductions and affirmative log surface cases

The results in this section concern smooth compactifications. Throughout, S is a smooth projective surface, $D \neq 0$ is a reduced simple normal crossing divisor, and $X = S \setminus \operatorname{supp} D$. Whenever a continuous plurisubharmonic exhaustion is assumed, it is finite-valued on X. The statements do not assert that an arbitrary singular log minimal endpoint has these smooth compactification properties.

6.1. Birational reductions and their scope.

The log surface MMP applies to projective pairs and, more generally, to suitable compact Kähler log surface pairs; see [11, 12]. A nef log-adjoint endpoint is controlled by $K_S + D$, whereas the exhaustion at infinity depends on the analytic neighborhood of D. These are distinct pieces of information.

The complement is preserved by a boundary contraction precisely when the upstairs removed set is the full inverse image of the downstairs removed set. An interior contraction with compact exceptional locus is covered by Proposition 5.7. General log MMP steps need not satisfy either condition.

Moreover, singular log endpoints may require a separate analysis of discrepancies and boundary coefficients on a resolution. We do not assume that arbitrary log MMP steps preserve either exhaustion property.

For a smooth projective pair with reduced SNC boundary, write $\overline{\kappa}(X) = \kappa(S, K_S + D)$ for its log Kodaira dimension. The log MMP leads to a Mori fiber space when this dimension is $-\infty$, and otherwise to a nef log-adjoint model. Log abundance then gives an associated morphism with image of dimension 0, 1, or 2. These correspond to a numerically trivial log adjoint, a curve Iitaka fibration, and log general type, respectively. For a smooth curve-fibration model with $(K_S + D) \cdot F = 0$, adjunction gives $2g(F) - 2 + D \cdot F = 0$; its general open fiber is therefore elliptic or $\mathbb{C}^*$. This organizes the cases below, but does not assert that passage to every log endpoint preserves the open surface. For a smooth irreducible boundary there is a useful restricted statement.

Proposition 6.1. *Let S be smooth projective and let $C \subset S$ be a smooth connected curve of genus at least two with $C^2 = 0$. Every birational $(K_S + C)$-negative extremal contraction from this smooth pair contracts a (-1)-curve disjoint from C. Thus a sequence of these birational steps restricts to finite proper modifications of $S \setminus C$, and both exhaustion properties are preserved.*

Proof. Adjunction gives $(K_S + C) \cdot C = 2g(C) - 2 > 0$, so C is not contracted. For an irreducible contracted curve $E \neq C$, one has $E^2 < 0$ and $C \cdot E \geq 0$. Again by adjunction,

$$(K_S + C) \cdot E = 2p_a(E) - 2 - E^2 + C \cdot E.$$

Negativity forces $p_a(E) = 0$, $E^2 = -1$, and $C \cdot E = 0$. The contraction is therefore the blowdown of a smooth rational (-1)-curve away from the boundary. After each such step the surface and the boundary remain smooth, with the same boundary genus and self-intersection. Proposition 5.7 applies at every step. □

Remark 6.2. Even when $K_S + D$ is semiample, its associated map need not be proper on $S \setminus \operatorname{supp} D$. Thus semiampleness of the log adjoint is not the assertion of Proposition 5.18, which requires semiampleness of a divisor supported on the boundary. The general remaining smoothing implication is not asserted in this article.

6.2. **Ueda-theoretic sufficient conditions for a smooth boundary.** We recall the neighborhood invariant used below. For a smooth curve C of square zero, give $N_{C/S}$ its unitary flat structure and choose local defining functions with first-order transition constants t_{ij} of modulus one. A system of type n satisfies

$$w_i = t_{ij} w_j + O(w_j^{n+1}).$$

The Ueda type is the largest attainable n, or infinity if such systems exist at every order. The successive obstructions lie in $H^1(C, N_{C/S}^{-n})$. Infinite type is a formal condition and does not by itself supply convergent defining functions.

Exact equations $w_i = t_{ij} w_j$ on a neighborhood give type (β); infinite type without such equations is type (γ) [13, §§2,5]. The Diophantine condition on N means that, for some $c, \tau > 0$ and an invariant distance on $\mathrm{Pic}^0(C)$,

$$\operatorname{dist}(N^{\otimes m}, \mathcal{O}_C) \geq c m^{-\tau} \qquad (m \geq 1).$$

Proposition 6.3 (A smooth square-zero boundary)**.** *Let C be a smooth connected curve on a smooth compact Kähler surface S, with $C^2 = 0$, and set $N = \mathcal{O}_C(C)$. The complement $S \setminus C$ admits a smooth psh exhaustion in each of the following cases:*

(i) *the Ueda type of $C \subset S$ is finite;*
(ii) *$\mathcal{O}_S(C)$ is unitary flat on a neighborhood of C;*
(iii) *N is torsion;*
(iv) *for some $k > 0$, the unitary representation associated with $N^{\otimes k}$ extends from $\pi_1(C)$ to $\pi_1(S)$;*
(v) *N satisfies the Diophantine condition;*
(vi) *C is elliptic.*

Proof. Case (i) follows from Ueda's strictly psh collar theorem [14, Theorem 2.5] and Lemma 5.11. In case (ii), unitary defining functions give the pluriharmonic collar function $-\log|w_i|^2$, to which the same lemma applies.

For (iii), finite Ueda type has already been treated. In infinite type, the fibration theorem for torsion normal bundle [14, Theorem 1.3] makes a positive multiple of C a fiber of a holomorphic fibration; a local base coordinate gives the required collar function.

For (iv), torsion normal bundles have already been treated in (iii), so only infinite Ueda type with non-torsion normal bundle remains. Neeman's extension theorem [14, Theorem 4.4] gives an effective divisor E disjoint from C with $[E] = m[C]$ in real cohomology for some $m > 0$. Hence $\mathcal{O}_S(mC - E)$ has zero real first Chern class and admits a unitary flat metric h by the $\partial\bar\partial$-lemma. If σ is its canonical meromorphic section, then $-m^{-1}\log\|\sigma\|_h^2$ is a smooth pluriharmonic collar diverging at C. This also gives the unitary flat neighborhood consequence. In (v), finite type is already covered, and Ueda's convergence theorem [13, Theorem 3] gives the flat neighborhood in infinite type. Finally, finite type again handles one alternative in (vi); for infinite type, [16, Theorem 1.6] excludes type (γ) and hence gives case (ii). □

6.3. **Homology and irregularity of a square-zero boundary.**

Lemma 6.4 (Exact character-extension criterion)**.** *Let $i : C \hookrightarrow S$ be the inclusion of a smooth connected compact complex curve in a compact complex surface, and let*

$$j = i_* : H_1(C, \mathbb{Z}) \longrightarrow H_1(S, \mathbb{Z}).$$

A character $\rho : H_1(C, \mathbb{Z}) \to \mathrm{U}(1)$ extends to $H_1(S, \mathbb{Z})$ if and only if it is trivial on $\ker j$. A positive power of ρ extends if and only if $\rho(\ker j)$ is finite. In particular, injectivity of j implies extension of every unitary character, without taking a power.

Proof. Necessity is immediate. If ρ is trivial on $\ker j$, define a character on the subgroup $j(H_1(C,\mathbb{Z}))$ by $j(a) \mapsto \rho(a)$. Every character from a subgroup of a finitely generated abelian group to $\mathrm{U}(1)$ extends to the whole group. Indeed, $\mathrm{U}(1)$ is divisible, and hence an injective abelian group. Alternatively one can use the structure theorem for finitely generated abelian groups and successively take roots in $\mathrm{U}(1)$. This argument allows torsion in $H_1(S,\mathbb{Z})$ and a nonprimitive image of j.

Apply the first assertion to ρ^k. Such a power extends exactly when $\rho(\ker j)$ is killed by a positive integer k, equivalently is contained in the finite group of kth roots of unity. Conversely every finite subgroup of $\mathrm{U}(1)$ is killed by its order. Finally, representations into $\mathrm{U}(1)$ factor through the abelianization, so the same statements hold for extension from $\pi_1(C)$ to $\pi_1(S)$. □

Theorem 6.5 (A homological sufficient condition)**.** *Let S be a smooth compact connected Kähler surface and $C \subset S$ a smooth connected curve with $C^2 = 0$. Let ρ be the unitary character associated with $N_{C/S}$, viewed as a character of $H_1(C,\mathbb{Z})$. If*

$$\rho\big(\ker(H_1(C,\mathbb{Z}) \to H_1(S,\mathbb{Z}))\big) \quad \textit{is finite,}$$

then $S \setminus C$ admits a smooth plurisubharmonic exhaustion. In particular this holds when $H_1(C,\mathbb{Z}) \to H_1(S,\mathbb{Z})$ is injective. No continuous exhaustion is assumed.

Proof. The degree-zero normal bundle is unitary flat. By Lemma 6.4, a positive power of its character extends to $H_1(S,\mathbb{Z})$ and hence to $\pi_1(S)$. Apply Proposition 6.3(iv). □

Corollary 6.6 (Irregularity at least the boundary genus)**.** *Let S and C be as in Theorem 6.5. If*

$$q(S) := h^1(S,\mathcal{O}_S) \;\geq\; g(C),$$

then $S \setminus C$ admits a smooth plurisubharmonic exhaustion. No continuous exhaustion is assumed.

Proof. Consider the restriction map

$$r : H^1(S,\mathcal{O}_S) \longrightarrow H^1(C,\mathcal{O}_C).$$

If r is not injective, [14, Proposition 2.7] implies that C is a multiple fiber of a holomorphic fibration $f : S \to B$. Taking connected fibers, $f^{-1}(b)$ has support exactly C for one point $b \in B$. Thus the restriction $S \setminus C \to B \setminus \{b\}$ is proper; pull back a smooth strictly subharmonic exhaustion of the punctured compact curve.

If r is injective, the inequality $q(S) \geq g(C)$ forces equality, and r is an isomorphism. The functorial Hodge decomposition on compact Kähler manifolds identifies r with the $(0,1)$ part of $i^* : H^1(S,\mathbb{C}) \to H^1(C,\mathbb{C})$; complex conjugation identifies its $(1,0)$ part. Therefore i^* is an isomorphism, and so is $i_* : H_1(C,\mathbb{Q}) \to H_1(S,\mathbb{Q})$.

The kernel of the integral map $j : H_1(C,\mathbb{Z}) \to H_1(S,\mathbb{Z})$ is consequently torsion. Since $H_1(C,\mathbb{Z}) \simeq \mathbb{Z}^{2g(C)}$ is torsion-free, that kernel is zero. Torsion in the target causes no difficulty: an element mapping to torsion maps to zero rationally, and hence must already be zero in the source. Apply Theorem 6.5; Lemma 6.4 shows that here the character itself extends, not merely a positive power. □

Remark 6.7 (The genus-zero endpoint). If $C \simeq \mathbf{P}^1$ and $C^2 = 0$, then $N_{C/S} \simeq \mathcal{O}_{\mathbf{P}^1}$. All Ueda obstruction groups $H^1(C, N_{C/S}^{-n})$ vanish, so the type is infinite. The torsion-normal-bundle fibration theorem gives a fibration with fiber C, and hence a smooth exhaustion of the complement. This is also covered by Corollary 6.6, since $q(S) \geq 0 = g(C)$.

6.4. **Smooth rulings in every boundary degree.** In a smooth ruling the square-zero alternative has a particularly rigid form. The following proposition includes affine-line complements and bisections, and removes any degree restriction.

Proposition 6.8 (Connected square-zero boundaries in a smooth ruling)**.** *Let $f : S \to B$ be a smooth $\mathbf{P}^1$-bundle over a connected smooth projective complex curve. Let $E = \bigcup_{i=1}^r E_i$ be a nonempty connected reduced simple normal crossing divisor. Suppose that there are integers $a_i > 0$ such that*

$$A = \sum_{i=1}^{r} a_i E_i, \qquad A \cdot E_i = 0 \quad (1 \leq i \leq r).$$

Then $S \setminus E$ admits a smooth plurisubharmonic exhaustion, without assuming a continuous exhaustion. More precisely, either E is one fiber, or $E = C$ is a single smooth curve with $C^2 = 0$ and $f|_C : C \to B$ finite étale. If its degree d is at least three, a finite étale base change identifies the pair with

$$(S', E') \simeq (B' \times \mathbf{P}^1,\ B' \times Q), \qquad |Q| = d.$$

Proof. Every vertical irreducible curve in S is a whole fiber, since the fibers are irreducible. If all components of E are vertical, their disjointness and connectedness of E make E one fiber. The map $S \setminus E \to B \setminus \{b\}$ is proper, so the pullback of a smooth subharmonic exhaustion of the punctured compact curve is an exhaustion. Such an exhaustion can, for example, be obtained from a metric of positive curvature on $\mathcal{O}_B(b)$ and the negative logarithm of the squared norm of its canonical section, followed by adding a constant.

Now suppose that E has a horizontal component. For a fiber class F, we have $A \cdot F > 0$. Thus no fiber can be a component of E, since $A \cdot E_i = 0$. All components are therefore horizontal. A $\mathbf{P}^1$-bundle over a smooth projective complex curve is the projectivization of a rank-two vector bundle; in particular S is projective and has a section C_0. Write

$$C_0^2 = -e, \qquad C_0 \cdot F = 1, \qquad F^2 = 0.$$

These classes form a basis of the real numerical divisor space. Put $a = A \cdot F > 0$ and write $A \equiv aC_0 + bF$. The orthogonality assumptions give

$A^2 = \sum_i a_i(A \cdot E_i) = 0$, so $b = ae/2$. Write $E_i \equiv d_i C_0 + b_i F$, where $d_i = E_i \cdot F > 0$. Then

$$0 = A \cdot E_i = a\big(b_i - d_i e/2\big), \qquad E_i \equiv d_i\big(C_0 + (e/2)F\big).$$

It follows that

$$E_i^2 = 0, \qquad E_i \cdot E_j = 0 \quad (i \neq j).$$

Distinct irreducible curves on a smooth surface have nonnegative local intersection multiplicities. Hence the E_i are pairwise disjoint; connectedness implies that $E = C$ is one component. The SNC hypothesis makes C smooth, and the preceding calculation gives $C^2 = 0$.

Set $d = C \cdot F$. The canonical numerical class is

$$K_S \equiv -2C_0 + (2g(B) - 2 - e)F, \qquad C \equiv d\big(C_0 + (e/2)F\big).$$

Consequently

$$K_S \cdot C = d(2g(B) - 2), \qquad 2g(C) - 2 = d(2g(B) - 2).$$

The second equality is adjunction. Since the nonconstant map $f|_C : C \to B$ is finite of degree d, Riemann–Hurwitz makes its ramification divisor zero. It is therefore étale for every d.

If $d = 1$, the complement is a holomorphic affine-line bundle over B. Its associated linear bundle has degree zero, because its dual is the normal bundle of the infinity section C, whose degree is $C^2 = 0$. We recall the affine-bundle argument, including the normal form needed for smoothness. Give the linear bundle a unitary flat metric and choose local affine coordinates with transitions $w_i = a_{ij} w_j + b_{ij}(z)$, where the a_{ij} are unitary constants. Choose a smooth section with local coordinates s_i. The forms $\bar\partial s_i$ give a global linear-bundle-valued $(0,1)$-form. Subtracting a global smooth section of the linear bundle makes this form harmonic. In a local flat frame over a curve, its coefficient is antiholomorphic. On sufficiently small coordinate disks choose antiholomorphic primitives g_i of this harmonic form. Then $s_i - g_i$ is holomorphic, and replacing w_i by $w_i - (s_i - g_i)$ gives transitions

$$w_i = a_{ij} w_j + c_{ij}, \qquad g_i = a_{ij} g_j + c_{ij}, \qquad |a_{ij}| = 1,$$

with constant c_{ij}: the differences are both holomorphic and antiholomorphic. This is the curve case of the normal form in [17, Proposition 2.1]. The functions $\rho_i = |w_i - g_i|^2$ agree globally. In coordinates (z, w_i),

$$(\rho_i)_{z\bar z} = |(g_i)_{\bar z}|^2, \qquad (\rho_i)_{w_i \bar w_i} = 1, \qquad (\rho_i)_{z\bar w_i} = (\rho_i)_{w_i \bar z} = 0.$$

Thus ρ is smooth and plurisubharmonic. Its sublevel sets are bundles of closed disks of bounded radius around a smooth section over compact B, and hence are compact.

For $d \geq 2$, choose a connected finite étale Galois cover $p : B' \to B$ killing the finite monodromy of $C \to B$. The pullback $S' = S \times_B B'$ has boundary E' consisting of d pairwise disjoint sections. If $q : S' \to S$ is the induced map and T is any of these sections, the projection formula gives

$$T^2 = (q^* C) \cdot T = C \cdot q_* T = 0,$$

since q^*C is the reduced union of these disjoint sections and q_*T is a positive integral multiple of C.

If $d = 2$, putting the sections at zero and infinity gives transition functions $w_i = \lambda_{ij}(z)w_j$. Thus $S' \setminus E' \simeq L^\times$ for a line bundle L of degree zero; the degree is, up to sign, the self-intersection of either section. For a unitary flat metric on L, the function $t = \log \|v\|^2$ is pluriharmonic on $L^\times$, and t^2 is a smooth nonnegative plurisubharmonic exhaustion.

If $d \geq 3$, use the unique fiberwise projective transformation putting three labeled sections at $0, 1, \infty$. It varies holomorphically on local trivializations; uniqueness makes the resulting trivializations agree. Hence $S' \simeq B' \times \mathbf{P}^1$. Every remaining section is a holomorphic map $B' \to \mathbf{P}^1$ avoiding $0, 1, \infty$. Such a map is constant, since a nonconstant map between compact connected curves is surjective. Thus $E' = B' \times Q$ for a fixed set of d points. Writing $Q = \{\infty, q_1, \ldots, q_{d-1}\}$, an explicit smooth nonnegative subharmonic exhaustion of $\mathbf{P}^1 \setminus Q$ is

$$r(z) = |z|^2 + \sum_{j=1}^{d-1} |z - q_j|^{-2}.$$

Its pullback to $B' \times (\mathbf{P}^1 \setminus Q)$ is proper because B' is compact.

In either case $d \geq 2$, let ρ' be the nonnegative exhaustion just constructed and let G be the deck group of $S' \setminus E' \to S \setminus C$. The average

$$\rho'_G = \frac{1}{|G|} \sum_{g \in G} \rho' \circ g$$

is smooth, plurisubharmonic, and invariant. It remains proper, because $\rho'_G \geq \rho'/|G|$. It descends smoothly through the finite étale cover to the required exhaustion of $S \setminus C$. □

Corollary 6.9 (The boundary-matrix criterion for smooth rulings)**.** *Let $f : S \to B$ be a smooth $\mathbf{P}^1$-bundle over a connected smooth projective complex curve, and let $D \neq 0$ be a reduced SNC divisor. The following conditions are equivalent:*

(i) *$S \setminus D$ admits a continuous plurisubharmonic exhaustion;*
(ii) *$S \setminus D$ admits a smooth plurisubharmonic exhaustion;*
(iii) *The intersection matrix of every connected component of D is not negative definite.*

In particular, there is no restriction on $D \cdot F$ or on the number of connected boundary components.

Proof. Smooth implies continuous, and Proposition 5.10 gives the necessary matrix condition. Conversely, consider one connected component $E = \bigcup_i E_i$ of D and its intersection matrix I. Its largest eigenvalue λ is nonnegative by hypothesis. The Perron–Frobenius argument in the proof of Proposition 5.13 uses only this matrix condition for the remaining conclusions. If $\lambda > 0$, it gives a positive integral divisor H supported on E with $H \cdot E_i > 0$, and hence a smooth strictly psh divergent collar by Lemma 5.12. If $\lambda = 0$, the

one-dimensional rational kernel has a strictly positive integral generator. Proposition 6.8 then supplies a smooth exhaustion of $S \setminus E$, whose restriction gives a collar at E. Lemma 5.11 combines the collars at all connected boundary components. □

Remark 6.10 (Scope). This criterion uses the rank-two numerical divisor space and the absence of reducible fibers of a smooth ruling. It does not settle a ruled surface after arbitrary blowups, a singular rational fibration, or the general projective square-zero boundary problem.

6.5. **Log Fano and log Calabi–Yau pairs.**

6.5.1. *The log Fano case.*

Proposition 6.11 (The smooth log Fano case)**.** *Suppose that $-(K_S + D)$ is ample. If X admits a continuous psh exhaustion, then it admits a smooth psh exhaustion.*

Proof. Apply Proposition 5.13 to each connected component Y of $\operatorname{supp} D$. The negative definite case is excluded by the continuous exhaustion hypothesis. A positive largest eigenvalue provides a smooth strictly psh collar function, which is sufficient by Lemma 5.11. In the remaining case there is an effective nef integral divisor A_Y, supported on Y, with $A_Y^2 = 0$ and $A_Y \cdot D_i = 0$ for every component of Y. The other connected boundary components are disjoint from Y, so $D \cdot A_Y = 0$. Therefore

$$K_S \cdot A_Y = (K_S + D) \cdot A_Y < 0.$$

Proposition 5.19 gives a smooth psh exhaustion of $S \setminus Y$, whose restriction is a collar function at Y. Choose disjoint collars for the finitely many connected boundary components and apply the construction of Lemma 5.11 separately on each collar. The sum of these functions is a smooth psh exhaustion of X. □

6.5.2. *The log Calabi–Yau case.*

Proposition 6.12 (Connected reduced boundary in the log Calabi–Yau case)**.** *Assume that D is connected and that $K_S + D \equiv 0$ numerically. If X admits a continuous psh exhaustion, then it admits a smooth psh exhaustion.*

Proof. For every irreducible component D_i, adjunction gives

$$0 = (K_S + D) \cdot D_i = 2g(D_i) - 2 + (D - D_i) \cdot D_i. \tag{6.1}$$

Since D is reduced SNC and connected, this implies that D is either a single smooth elliptic curve or a cycle of smooth rational curves. Here a cycle with two components has two transverse intersection points.

The negative definite and positive largest-eigenvalue cases of the boundary intersection matrix are treated by Proposition 5.13. It remains to consider the negative semidefinite case with a strictly positive kernel vector. For a single elliptic component this means $D^2 = 0$. If its Ueda type is finite, Ueda's theorem provides a smooth strictly psh collar function tending to infinity at

D [14, Theorem 2.5]. If its Ueda type is infinite, the exclusion of type (γ) for elliptic curves in compact complex surfaces [16, Theorem 1.6] gives a unitary flat neighborhood. In the latter neighborhood, unitary defining functions $w_i = t_{ij} w_j$, $|t_{ij}| = 1$, give the smooth pluriharmonic function $-\log |w_i|^2$. Lemma 5.11 concludes the elliptic case.

Now suppose that D is a rational cycle. Contract any component which is a (-1)-curve. This is the inverse of blowing up a boundary node and taking the full reduced total transform. If $q : S \to S'$ is this contraction and D' is the image cycle, then

$$q^* D' = D + E, \qquad K_S = q^* K_{S'} + E, \qquad K_S + D = q^*(K_{S'} + D').$$

Thus $S \setminus D \simeq S' \setminus D'$ and $K_{S'} + D' \equiv 0$. Contracting a component of a three-component cycle yields a two-component cycle; contracting a component of a two-component cycle yields an irreducible nodal rational curve. Every intermediate support is a reduced nodal cycle.

The existence of a positive square-zero weighted divisor is also preserved. Indeed, a nef kernel divisor has intersection zero with the contracted curve and hence is the numerical pullback of its pushforward. Stop when the boundary has no smooth (-1)-component, or when it has become irreducible nodal. In the irreducible case, the existence of a nonzero square-zero weighted divisor implies $D^2 = 0$.

In the remaining reducible case write $b_i = -D_i^2$ and let $A = \sum a_i D_i$, $a_i > 0$, be a kernel divisor. Negative semidefiniteness, positive intersections with the adjacent components, and the absence of (-1)-components imply $b_i \geq 2$. Adding the equations $A \cdot D_i = 0$ gives

$$\sum_i (b_i - 2) a_i = 0,$$

where the two edges incident to a component are counted separately also for a two-component cycle. Hence $b_i = 2$ for all i. The kernel equations on a connected cycle then force all a_i to be equal. Consequently the reduced divisor D itself is nef and $D^2 = 0$.

The resulting surface is rational. Indeed, $K_S \equiv -D$ implies $K_S \cdot H < 0$ for an ample divisor H, so $\kappa(S) = -\infty$. If S were nonrational ruled, its rational curves would be vertical for the ruling over a curve of positive genus. The supports of fibers of such a ruling, including after point blowups, have tree dual graphs; they contain neither a rational cycle nor an irreducible nodal rational curve. The Enriques–Kodaira classification therefore makes S rational. On a rational surface numerical triviality of a line bundle implies its triviality, so

$$\mathcal{O}_S(D) \simeq K_S^{-1}.$$

The normal line bundle $N_D = \mathcal{O}_D(D)$ has multidegree zero. Since D is a rational nodal cycle, $\mathrm{Pic}^0(D) \simeq \mathbb{C}^*$. Let $q_D \in \mathbb{C}^*$ be its period. Changing the orientation of the cycle replaces q_D by q_D^{-1}, so the following alternatives do not depend on that choice.

If $|q_D| = 1$, the semipositivity theorem for generalized Halphen surfaces gives a smooth semipositive Hermitian metric h on $\mathcal{O}_S(D)$ [16, Proposition 6.2(B)]. The function $-\log \|s_D\|_h^2$ on $S \setminus D$ is therefore a smooth psh exhaustion.

If $|q_D| \neq 1$, then every N_D^{-m} is a nontrivial multidegree-zero line bundle on the rational cycle. The normalization exact sequence gives

$$H^0(D, N_D^{-m}) = H^1(D, N_D^{-m}) = 0 \qquad (m \geq 1).$$

In particular, all the nodal Ueda obstruction classes vanish, so the pair has infinite type. Koike's theorem for nonunitary normal periods [15, Theorem 1.6] then supplies a smooth strictly psh function on a punctured neighborhood, tending to infinity at D. Lemma 5.11 completes this last case. □

Remark 6.13. The elliptic infinite-type step and the unitary-period rational-cycle step above use the preprint [16], submitted in September 2026. They are stated here with that status. The proof does not identify finite Ueda type with smooth semipositivity of $\mathcal{O}_S(D)$: in that case it uses the collar function supplied by Ueda's theorem.

6.6. **Elliptic fibrations and vertical boundaries.**

Proposition 6.14 (The vertical-boundary case)**.** *Let $f : S \to B$ be a morphism with connected fibers onto a smooth projective curve. Suppose that its general fiber F is a smooth elliptic curve and $(K_S + D) \cdot F = 0$. If X admits a continuous psh exhaustion, then there is a finite nonempty set $P \subset B$ such that*

$$\operatorname{supp} D = f^{-1}(P), \qquad X = f^{-1}(B \setminus P).$$

In particular, X admits a smooth psh exhaustion. This applies when f is the log Iitaka fibration with elliptic general fiber.

Proof. Since $F^2 = 0$ and $g(F) = 1$, adjunction gives $K_S \cdot F = 0$. Thus $D \cdot F = 0$, and every irreducible component of D is vertical. Only finitely many fibers meet the boundary.

For a fiber $F_b = \sum_i m_i C_i$, the intersection identity (5.1) shows that every proper principal submatrix is negative definite: its vector extends by zero on an omitted component and therefore cannot lie in the kernel spanned by the positive multiplicity vector.

If the boundary met F_b without containing its whole support, each of its connected boundary subconfigurations in that fiber would have negative definite intersection matrix. Such a connected component is prohibited by Proposition 5.13. Consequently the boundary contains whole fiber supports. Multiple fibers cause no change, since the conclusion concerns supports. The restricted map $f|_X : X \to B \setminus P$ is proper. Pull back a smooth strictly subharmonic exhaustion of the punctured base. □

7. Examples and remaining boundary geometry

7.1. **Smooth exhaustions without holomorphic convexity.** The log general type condition does not force holomorphic convexity of the complement. The following affirmative examples also show that ramification of the boundary over an Iitaka base is compatible with a smooth exhaustion.

Proposition 7.1. *Let B be a smooth projective curve of genus at least two, and let $L \in \mathrm{Pic}^0(B)$ be non-torsion. Put*

$$S_0 = \mathbf{P}_{\mathrm{lines}}(\mathcal{O}_B \oplus L), \qquad \pi : S_0 \to B, \qquad C_\infty = \mathbf{P}_{\mathrm{lines}}(L).$$

For an integer $n \geq 1$, choose a line bundle A of sufficiently large degree that the divisor class $H_n = nC_\infty + \pi^* A$ is very ample. Choose a smooth branch divisor $R \in |2H_n|$ transverse to C_∞, let $q : T \to S_0$ be its double cover, and put $D = q^{-1}(C_\infty)$. Then T is smooth projective, D is smooth connected with $D^2 = 0$, and $X = T \setminus D$ admits a smooth psh exhaustion but satisfies $\mathcal{O}(X) = \mathbb{C}$. Moreover,

$$\kappa(T, D) = 0, \qquad \kappa(T, K_T + D) = \begin{cases} 1, & n = 1, \\ 2, & n \geq 2. \end{cases}$$

For $n \geq 3$, K_T itself is ample.

Proof. The curve C_∞ has normal bundle L^{-1} and square zero; it is nef. The ruled-surface canonical bundle formula, with our lines convention, is

$$K_{S_0} \sim -2C_\infty + \pi^*(K_B \otimes L^{-1}).$$

The double cover is defined by the algebra $\mathcal{O}_{S_0} \oplus \mathcal{O}_{S_0}(-H_n)$ and the chosen section cutting out R. Smoothness of R makes T smooth, and transversality makes D smooth. The restriction $D \to C_\infty$ is a double cover branched at $R \cap C_\infty$, which has $2 \deg A > 0$ points. This implies connectedness of D. The projection formula gives $D^2 = 2C_\infty^2 = 0$.

The double-cover canonical bundle formula gives

$$K_T + D \sim q^*\big((n-1)C_\infty + \pi^*(K_B \otimes L^{-1} \otimes A)\big),$$
$$K_T \sim q^*\big((n-2)C_\infty + \pi^*(K_B \otimes L^{-1} \otimes A)\big).$$

The base line bundle in these formulas has positive degree. For $n = 1$, the log adjoint is pulled back from this ample base bundle and has Iitaka dimension one. For $n \geq 2$, the first divisor is ample: it has positive intersection with every curve and positive square, by nefness of C_∞ and the ruled-surface intersection formulas. Nakai–Moishezon and finite pullback give ampleness of $K_T + D$. The same argument applies to K_T for $n \geq 3$.

There is an identification $S_0 \setminus C_\infty = \mathrm{Tot}(L)$. Equip L with its unitary flat metric. The squared norm $\rho_0(v) = \|v\|^2$ is a smooth psh exhaustion of this total space. The finite map $q|_X$ is proper, so

$$\rho = \rho_0 \circ q|_X$$

is a smooth psh exhaustion of X.

To check the absence of holomorphic functions, expand a holomorphic function on $\mathrm{Tot}(L)$ in fiber Taylor series. The coefficient of degree j is a global section of L^{-j}. For $j > 0$ this bundle is nontrivial of degree zero and has no section. The degree-zero coefficient is constant on B. Therefore $\mathcal{O}(\mathrm{Tot}(L)) = \mathbb{C}$. If $f \in \mathcal{O}(X)$, its trace and norm under the finite double cover are holomorphic functions on $\mathrm{Tot}(L)$, hence constants a and b. Thus $f^2 - af + b = 0$. Connectedness of X forces f to be constant.

Finally, the ratio of any section of $\mathcal{O}_T(mD)$ to the canonical section s_D^m is holomorphic on X. The preceding argument implies that it is constant. Hence $h^0(T, \mathcal{O}_T(mD)) = 1$ for all $m \geq 1$, and $\kappa(T, D) = 0$. □

Remark 7.2. For $n = 1$, a general fiber of $T \to B$ is a double cover of $\mathbf{P}^1$ branched at two points, hence is rational; removing D gives $\mathbb{C}^*$. Nevertheless, $D \to B$ is ramified at $R \cap C_\infty$. Thus a proof of the general rational-fiber Iitaka case cannot assume that its boundary multisection is unramified merely because a smooth psh exhaustion exists. For $n \geq 3$ these examples have an ample ordinary canonical bundle as well as an ample log adjoint, while their complements are not holomorphically convex. They are examples with smooth exhaustions, not counterexamples to any continuous-to-smooth implication.

7.2. **Square-zero boundaries with prescribed normal bundle.** The preceding propositions do not prove the smoothing implication for every log Iitaka pair with rational general fiber, nor for every log general type pair. The following constructions explain why log Kodaira dimension alone does not make the residual boundary geometry finite or automatic. They are numerical test families, not counterexamples to the smoothing implication; no continuous exhaustion is asserted for arbitrary choices of their parameters.

Proposition 7.3 (Square-zero boundaries with arbitrary normal bundle)**.** *Let $S_0 = \mathbf{P}^1 \times \mathbf{P}^1$, and let $C \subset S_0$ be a smooth curve of bidegree (a, b), where $a, b \geq 2$. For every $N \in \mathrm{Pic}^0(C)$ there are $2ab$ distinct points on C such that, after blowing them up and taking the strict transform D of C, one has*

$$D^2 = 0, \qquad N_{D/S} \simeq N, \qquad D \text{ is nef},$$
$$\mathcal{O}_S(K_S + D) \simeq \pi^* \mathcal{O}_{S_0}(a - 2, b - 2).$$

If N is non-torsion, then $\kappa(S, D) = 0$. If $a = 2$, $b \geq 3$, the pair is log minimal of log Kodaira dimension one. If $a, b \geq 3$, it is log minimal of log Kodaira dimension two.

Proof. The genus and self-intersection are

$$g(C) = (a - 1)(b - 1), \qquad C^2 = 2ab.$$

The line bundle $\mathcal{O}_C(C) \otimes N^{-1}$ has degree $2ab$, which exceeds $2g(C)$. It is therefore basepoint-free and a general section has $2ab$ distinct zeros. Let Z be this reduced divisor of zeros and let $\pi : S \to S_0$ be the blowup of its

points. The normal-bundle formula for a strict transform gives

$$N_{D/S} \simeq \mathcal{O}_C(C - Z) \simeq N,$$

and $D^2 = C^2 - \deg Z = 0$. Since D is irreducible, its intersections with distinct curves are nonnegative; together with $D^2 = 0$, this proves nefness.

Write E_j for the exceptional curves. The blowup formulas give

$$K_S = \pi^* K_{S_0} + \sum_j E_j, \qquad D = \pi^* C - \sum_j E_j.$$

Thus $K_S + D \sim \pi^*(K_{S_0} + C)$, proving the displayed formula. For $(a, b) = (2, b)$ with $b \geq 3$, this is the pullback of $\mathcal{O}_{\mathbf{P}^1}(b-2)$ under the second projection. For $a, b \geq 3$ the corresponding line bundle on S_0 is ample. In both cases its pullback is nef and semiample, with the claimed Iitaka dimension.

If N is non-torsion, then $H^0(D, N^{\otimes m}) = 0$ for every $m > 0$. Induction in the sequences

$$0 \longrightarrow \mathcal{O}_S((m-1)D) \longrightarrow \mathcal{O}_S(mD) \longrightarrow N^{\otimes m} \longrightarrow 0$$

gives $h^0(S, \mathcal{O}_S(mD)) = 1$ for all $m \geq 0$. This is exactly $\kappa(S, D) = 0$. □

Remark 7.4 (Crepant boundary-point blowups). The exceptional curves in Proposition 7.3 satisfy $(K_S + D) \cdot E_j = 0$. Thus they are not removed by a negative log extremal contraction. Moreover $E_j \setminus D \simeq \mathbb{C}$ belongs to the open complement. Blowing it down sends this open curve into a point of the removed boundary and is not a proper modification of $S \setminus D$ onto $S_0 \setminus C$. This distinguishes these blowups from boundary-node blowups with full reduced total transform, which preserve the open complement.

8. Holonomy obstructions and the remaining surface problem

The previous sections give complete criteria in the nonprojective and smooth ruled settings, as well as further affirmative projective cases. We now isolate what the neighborhood arguments imply, and what is still missing for an arbitrary continuously weakly complete surface.

8.1. **Two obstructions for foliated neighborhoods.**

The statements below concern the indicated foliated or suspension neighborhoods. They neither assert that an arbitrary plurisubharmonic collar is basic nor produce a foliation from the existence of such a collar.

Proposition 8.1 (A continuous basic collar linearizes holonomy)**.** *Let C be a compact leaf of a nonsingular holomorphic foliation on a complex surface, defined on a neighborhood U of C. Suppose that a finite continuous function $u : U \setminus C \to \mathbb{R}$ is constant on leaves and tends to $+\infty$ as the point approaches C. Then the holonomy along C is simultaneously holomorphically conjugate to a group of rotations. On a possibly smaller neighborhood V, the divisor bundle $\mathcal{O}_V(C)$ is unitary flat. In particular, $V \setminus C$ admits a smooth pluriharmonic function tending to $+\infty$ along C.*

Proof. Fix a transverse coordinate w with $C = \{w = 0\}$, and put $v = u|_{\{w\neq 0\}}$. Take representatives of finitely many holonomy generators and their inverses. Choose $0 < r < R$ so that these maps are univalent on a neighborhood of $\overline{B_R}$, carry B_r into B_R, and preserve v wherever used. The closed disk $\overline{B_R}$ is chosen inside the transverse chart. Take

$$t > \max_{r\leq|w|\leq R} v(w), \qquad E = \{0\} \cup \{w \in B_R \setminus \{0\} : v(w) > t\}.$$

The divergence of v makes E open at 0, and $E \Subset B_r$. Invariance of v gives $h(E) \subset E$ and $h^{-1}(E) \subset E$, hence $h(E) = E$, for every chosen generator h. The component D of E containing 0 is consequently a bounded domain preserved by all these maps.

The isotropy group $G = \mathrm{Aut}(D, 0)$ is compact in the compact-open topology. This standard normal-family fact can be seen by applying Cauchy's estimate to all iterates and their inverses, which gives $|g'(0)| = 1$, and then taking simultaneous locally uniform limits of automorphisms and their inverses. Let μ be normalized Haar measure on G and define

$$\psi(w) = \int_G g'(0)^{-1} g(w)\, d\mu(g).$$

This is holomorphic on D, satisfies $\psi(0) = 0$ and $\psi'(0) = 1$, and right invariance of Haar measure gives

$$\psi(h(w)) = h'(0)\psi(w) \qquad (h \in G).$$

Thus ψ is a local simultaneous linearizing coordinate and the multipliers are unitary. No filling of complementary components of D is required.

Transport this coordinate along finitely many flow boxes covering C. Path changes give holonomy germs, so, after shrinking, the resulting holomorphic first integrals w_i satisfy $w_i = \lambda_{ij} w_j$ on connected overlaps, with constant $|\lambda_{ij}| = 1$. They are local defining functions for C. Their ratios therefore give unitary-constant transition functions for $\mathcal{O}_V(C)$ and define a smooth flat Hermitian metric on that bundle. The functions $-\log|w_i|$ agree, are smooth and pluriharmonic off C, and diverge along C. □

Proposition 8.2 (Cyclic nonlinearizable suspensions have no psh collar)**.** *Let C be a compact connected Riemann surface of genus at least one, let $\chi : \pi_1(C) \to \mathbb{Z}$ be primitive, and let f be a nonlinearizable irrationally indifferent holomorphic diffeomorphism germ at 0. Suppose that the germ of a neighborhood of C is the holomorphic suspension of $\gamma \mapsto f^{\chi(\gamma)}$. There is no finite continuous plurisubharmonic function on a punctured neighborhood of C tending to $+\infty$ along C.*

Proof. Choose an oriented nonseparating smooth simple closed curve a whose Poincaré dual is χ. Cutting along a gives a connected compact bordered surface B with two boundary circles. The character χ vanishes on $\pi_1(B)$, and the cyclic cover C_χ is obtained by chaining copies B_n, $n \in \mathbb{Z}$. The suspension is trivial over B; its two sides are glued using f.

We use the following precise dynamical input: in arbitrarily small admissible transverse neighborhoods there is a nontrivial compact connected completely invariant hedgehog K, with $0 \in \partial K$, and a point $x \in \partial K$ whose forward orbit is dense in ∂K. The existence and two-sided density are recalled in [18, Theorems 2.1 and 2.3]; the passage to forward density uses Pérez-Marco's recurrence theorem, as explicitly explained in [18, Section 5]. Forward density is not being inferred from two-sided density alone.

Choose K small enough that its compact suspension is contained in the neighborhood on which the proposed function u is defined off C. The point x is nonperiodic, since its orbit closure contains both 0 and other points. The leaf through x is therefore biholomorphic to C_χ. Write $w_n = f^n(x)$. Choose the boundary-coordinate convention so that the cut preceding B_n has parameter w_n, and the interior of B_n is the copy of B with transverse parameter w_n.

Fix $y \in \partial K \setminus \{0\}$. The lifted cut circle with parameter y is compact and disjoint from C. Continuity gives a transverse neighborhood V_y of y and a real number M such that $u < M$ on all cut circles with parameter in V_y. Fix a point b in the interior of B. Divergence along C gives a neighborhood V_0 of 0 for which $u(b, w) > M$ whenever $w \in V_0 \setminus \{0\}$. Every tail of the forward dense orbit is dense: ∂K has no isolated points, so deleting finitely many orbit points cannot destroy density. We may consequently choose

$$n_1 < n_2 < n_3, \qquad w_{n_1}, w_{n_3} \in V_y, \qquad w_{n_2} \in V_0.$$

The chain $B_{n_1} \cup \cdots \cup B_{n_3 - 1}$ is a compact connected bordered Riemann surface in this leaf. Its only boundary components are the cuts with parameters w_{n_1} and w_{n_3}, where $u < M$; the interior point (b, w_{n_2}) has $u > M$. The restriction of u to the leaf is subharmonic, contradicting the maximum principle. □

Remark 8.3 (Scope). Subharmonicity is unnecessary in Proposition 8.1, but leafwise constancy is essential to its argument. Proposition 8.2 does not require leafwise constancy or a parabolic universal cover; it does require the specified holomorphic cyclic suspension. Neither proposition upgrades an arbitrary continuous psh collar to a basic one or supplies a foliation or holomorphic suspension for a general surface neighborhood. Consequently these obstructions do not settle the unrestricted continuous-to-smooth implication.

Remark 8.4 (Pseudoconvex neighborhoods are a different condition). Koike–Takakura [19, Theorem 1.1] consider a holomorphic section of a holomorphic submersion with one-dimensional fibers over a compact complex manifold, with unitary flat normal bundle. They prove that a fundamental system of locally pseudoconvex neighborhoods is equivalent to a holomorphic tubular neighborhood. The submersion and the pseudoconvex neighborhood basis are hypotheses of that result, not consequences of the collar assumption here. In particular, a smooth psh function u tending to $+\infty$ along C gives, at regular levels, neighborhoods on the side $\{u > t\}$ with local defining function $t - u$. Its Levi form on complex tangent directions is nonpositive. Thus a

divergent psh function on the *punctured* neighborhood does not by itself provide the pseudoconvex neighborhood basis required in that theorem. We do not use it to close the remaining surface argument.

8.2. **Necessary conditions in the smooth-boundary case.**

Corollary 8.5 (Necessary conditions for a smooth-boundary counterexample)**.** *Suppose that S is a smooth projective surface, $C \subset S$ is a smooth connected curve with $C^2 = 0$, and $X = S \setminus C$ has a continuous plurisubharmonic exhaustion but no smooth one. Then*

$$\mathcal{O}(X) = \mathbb{C}, \qquad q(S) < g(C),$$

and

$$h^0(S, \mathcal{O}_S(mC)) = 1 \qquad (m \geq 1),$$

and the unitary character of $N_{C/S}$ has infinite image on $K = \ker(H_1(C, \mathbb{Z}) \to H_1(S, \mathbb{Z}))$. Moreover,

$$\operatorname{rank} K = 2\big(g(C) - q(S)\big) \geq 2.$$

The Ueda type is infinite, the neighborhood is not unitary flat, and $N_{C/S}$ is non-torsion and fails the Diophantine condition in Proposition 6.3. Using the September 2026 preprint [16, Theorem 1.6]*, one further has $g(C) \geq 2$. In addition, every continuous psh exhaustion u of X satisfies $(a_u, +\infty) \subset u(\Sigma^0_X)$ for some $a_u \in \mathbb{R}$.*

Proof. Proposition 5.4 gives $\mathcal{O}(X) = \mathbb{C}$. For every section s of $\mathcal{O}_S(mC)$, the quotient s/s_C^m is holomorphic on X, hence constant. The canonical section s_C^m is nonzero, so $h^0(S, \mathcal{O}_S(mC)) = 1$. Corollary 6.6 and Theorem 6.5 give the homological conditions. If the restriction $H^1(S, \mathcal{O}_S) \to H^1(C, \mathcal{O}_C)$ were not injective, Proposition 2.7 of [14] would make C a multiple fiber, again giving a smooth exhaustion. It is therefore injective. Hodge decomposition and duality make $H_1(C, \mathbb{Q}) \to H_1(S, \mathbb{Q})$ surjective. Their dimensions are $2g(C)$ and $2q(S)$, respectively. Tensoring the integral kernel sequence with $\mathbb{Q}$ proves the stated rank formula, even when the target has torsion. The remaining exclusions follow from Proposition 6.3. The minimal-kernel assertion is Proposition 5.5. The elliptic exclusion uses the stated 2026 preprint; the homological and irregularity criteria do not depend on that preprint. □

8.3. **The remaining analytic problem.**

Remark 8.6 (The remaining smooth-boundary subproblem)*.* Even when C is a smooth connected curve in a smooth projective surface S, with $C^2 = 0$ and $g(C) \geq 2$, the following implication has not been proved here:

$$u \in C^0(U \setminus C) \cap \mathrm{PSH}(U \setminus C), \qquad \lim_{x \to C} u(x) = +\infty$$
$$\Longrightarrow \quad \exists\, \Phi \in C^\infty(V \setminus C) \cap \mathrm{PSH}(V \setminus C), \qquad \lim_{x \to C} \Phi(x) = +\infty,$$

where U and V are neighborhoods of C. A smooth collar would extend to a smooth exhaustion by Lemma 5.11. Proposition 6.3 settles finite Ueda

type, unitary flat neighborhoods, torsion or Diophantine normal bundles, the stated monodromy-extension condition, and elliptic boundary curves. The homological and irregularity criteria sharpen this class as in Corollary 8.5. The remaining smooth-boundary situation has infinite Ueda type and non-torsion normal monodromy with infinite image on the homology kernel. Knowing only the degree-zero normal bundle does not settle it. The foliated results above do not remove the gap: an arbitrary continuous psh collar need not have been shown to be basic, and a holomorphic suspension is additional geometric structure. This article does not prove the unrestricted continuous-to-smooth implication for surfaces.

Reducible weighted square-zero boundaries and singular log endpoints with changed boundary coefficients require additional arguments as well. Semiampleness of $K_S + D$ provides a map of the compact pair, but does not ensure that its restriction to the divisor complement is proper.

In particular, an almost-positive regularization estimate is insufficient on its own. If $i\partial\bar\partial u_\varepsilon \geq -\varepsilon\omega$, then

$$i\partial\bar\partial\chi(u_\varepsilon) = \chi'(u_\varepsilon)i\partial\bar\partial u_\varepsilon + \chi''(u_\varepsilon)i\partial u_\varepsilon \wedge \bar\partial u_\varepsilon.$$

The second term controls only one complex direction. On $\ker \partial u_\varepsilon$, a negative Levi direction can remain. Therefore convex composition alone does not turn such estimates into the missing semipositive exhaustion.

8.4. **Conclusion and dependence on recent results.** The threefold counterexamples distinguish continuous weak completeness from C^2 weak completeness, including in the quasi-projective Kähler category and after a modification supported over one point. In dimension two, the numerical criteria completely settle divisor complements with nonprojective compactifier and reduced SNC complements in a smooth ruling. The projective residual problem still includes connected weighted square-zero boundaries and the smooth-boundary class in Corollary 8.5.

The elliptic and unitary-period rational-cycle arguments explicitly use [16], a preprint submitted September 22, 2026. Section 1.3.1 of that preprint records the construction of a type (γ) embedding in a compact complex surface as an open problem, and Theorem 1.6 excludes elliptic curves. A smooth-boundary counterexample in our residual class would give such a compact type (γ) embedding. The converse is not established: such an embedding alone would give neither a continuous psh exhaustion of its complement nor a proof that every smooth psh exhaustion is impossible. Neither the homological criterion nor the all-degree smooth-ruling criterion depends on this preprint.

Finally, weak completeness has not supplied a divisorial compactification of an arbitrary noncompact surface. Accordingly the full dimension-two question remains unresolved by the arguments here, both without further assumptions and under the sole additional assumption of Kählerness or quasi-projectivity.

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